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Impact of Anthropogenic Emission Estimates on Air Quality and Human Health Effects

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Abstract Global bottom-up anthropogenic emission inventories show substantial spatial and temporal differences of short-lived pollutant emissions, which results in uncertainties in terms of air quality and human health impacts. In this study, we compare the emissions of trace gases and aerosols for the year 2015 from three different global emission inventories, the Community Emissions Data System (CEDS), the Copernicus Atmosphere Monitoring Service Global Anthropogenic Emissions (CAMS-GLOB-ANT), and Evaluating the Climate and Air Quality Impacts of Short-Lived Pollutants version 6b (ECLIPSEv6b). We then employ the Community Atmosphere Model with chemistry version 6.0 within the Community Earth System Model version 2.2.0 to quantify the atmospheric chemistry and air quality impacts from the above three anthropogenic emission inventories, with a focus on $PM_{2.5}$ (particulate matter with aerodynamic diameters equal or less than 2.5 μm) and ozone (O_3). Our results indicate that differences between emission inventories are largest for black carbon, organic carbon, ammonia and sulfur dioxide, in terms of global annual total emissions. These differences in emissions across CEDS, CAMS, and ECLIPSEv6b lead to substantial variations in global annual totals and spatial distribution patterns. This study shows that the global annual total $PM_{2.5}$-induced premature mortality is three times higher than that from O_3 mortality, indicating that $PM_{2.5}$ is the primary contributor compared with O_3. An inter-comparison of global human health impacts from CEDS, CAMS and ECLIPSEv6b indicates that 80% (CEDS), 81.2% (CAMS), and 77.6% (ECLIPSEv6b) of premature deaths due to anthropogenic activities are associated with Asia and Africa continents.

Plain Language Summary In this study, we analyze the differences in emissions of pollutants from three commonly used global anthropogenic emission inventories for the year 2015. We find that there exist large differences among these data sets in terms of global annual total emissions and spatial distributions. By simulating the impact of these emissions on atmospheric chemistry and air quality through a global model, the differences in emissions lead to variations in estimates of particulate matter and ozone levels and their health impacts. Our results show that particulate matter is responsible for three times more premature deaths globally than ozone, with a significant portion of the deaths concentrated in Asia and Africa. This comparison highlights the importance of reducing uncertainties in emission inventories to improve estimates of air pollution and its health impacts.

Key Points:

- This study compared three global anthropogenic emission inventories and their associated air quality and human health impacts
- Large discrepancies of annual total emissions were found among the three anthropogenic emission inventories
- Over 80% of global annual total $PM_{2.5}$- and ozone-induced premature deaths from anthropogenic sources occurred in Asia and Africa

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

Air pollution plays a major role in the deterioration of our ambient air quality and human health as well as driving the changes of our climate associated with emissions of greenhouse gases and short-lived climate forcers (SLCFs) (McDuffie et al., 2020). According to the Sixth Assessment Report (AR6) of the Intergovernmental Panel on Climate Change (IPCC), a United Nation organization for assessing the science related to climate change, anthropogenic induced climate change has already impacted weather and climate extremes globally (Huang et al., 2018, 2021; IPCC, 2023; Lund et al., 2023).

Emissions of SLCFs associated with human activities include black carbon (BC), organic carbon (OC), nitrogen oxides (NO_x = NO + NO_2), ammonia (NH_3), carbon monoxide (CO), sulfur dioxide (SO_2), and non-methane volatile organic compounds (NMVOCs), which have altered the composition of the atmosphere and the related fluxes from land and oceanic surfaces (Hoesly et al., 2018; Lund et al., 2020).

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Anthropogenic emissions inventories, comprised of comprehensive databases for pollutants and greenhouse gases, are critical for understanding the emissions' impact on the environment and human health using chemistry-climate models (McDuffie et al., 2020). However, there are limits and discrepancies in emission inventories, which result in uncertainties in quantifying the impact of anthropogenic emissions of chemicals and aerosol precursors on human health. Shami et al. (2022) suggests that emission inventories such as Emissions Database for Global Atmospheric Research (EDGAR), although applicable for continental scale studies, are deemed to having an underestimation when compared to ground observations. Additionally, a significant gap exists in inventories due to the lengthy process of collecting data from all countries, leading to delays in reflecting the most current information in updates (Elguindi et al., 2020; Lund et al., 2023). Difference in process methodology and the data set's coverage, which includes the geographic range and type of emission activities, can also contribute to differences between inventories (Andrew, 2020; Elguindi et al., 2020).

Past studies have used anthropogenic emissions inventories, the Community Emissions Data System (CEDS), the Copernicus Atmosphere Monitoring Service (CAMS) and Evaluating the Climate and Air Quality of Short-lived Pollutants version 6b (ECLIPSEv6b, hereafter referred to as ECLIPSE) for air quality modeling and analysis. Along with human health impact assessments, emission inventories have been employed to estimate the distribution and magnitude of air pollutants across different regions (Gaubert et al., 2021; Huang et al., 2018, 2021; Jo et al., 2023; Kurokawa & Ohara, 2020; Nawaz et al., 2023; Pino-Cortés, 2021; Upadhyay et al., 2020; Xiong, Partha, et al., 2022; Xiong et al., 2024). Intercomparison studies have been conducted to analyze the differences between emission inventories; Elguindi et al. (2020) compared top-down and bottom-up emissions inventories, finding that annual emissions for species such as CO, BC, and OC were highest in CEDS and lowest in EDGAR for regions like China and India across various pollutants. Kurokawa and Ohara (2020) compared between CEDS and EDGAR, highlighting differences in emissions estimated for pollutants such as NO_x, CO and $PM_{2.5}$. This is relevant as CAMS is a combination of CEDS and EDGAR, along with other emission data sets. Choulga et al. (2020) compared the global uncertainties between CEDS and EDGAR, where EDGAR's global uncertainty was quantified and CEDS's was not. We would also like to note that ECLIPSE is used by both the CAMS and CEDS emissions inventories. CEDS incorporates ECLIPSE to refine emissions from sectors like gas flaring and shipping (Hoesly et al., 2018), ensuring more accurate data. Similarly, CAMS integrates ECLIPSE (Granier, et al., 2019) to enhance the accuracy of emissions estimates related to specific sectors, such as agricultural waste burning. In this study, we compare three commonly used global emission inventories, including CEDS, CAMS, and ECLIPSE.

Anthropogenic air pollution poses a significant threat to public health, and epidemiological studies have found links between outdoor air pollution and adverse health outcomes, including morbidity and premature deaths associated with respiratory and cardiovascular diseases (e.g., stroke and lung cancer) (Dong et al., 2012; Huang et al., 2020; Lelieveld et al., 2015; Pope, 2000; Turner et al., 2016). It has been estimated that seven million people die each year due to indoor and outdoor air pollution (Sarkodie et al., 2019; World Health Organization, 2024). Even at low exposure levels, long-term exposure has a considerably greater effect on public health (Beverland et al., 2012; Pozzer et al., 2023). Past studies have reported that global annual premature deaths attributable to ambient long-term $PM_{2.5}$ (particulate matter with aerodynamic diameters $\leq$2.5 μm) exposure were estimated to be 4 million in 2015 (Burnett et al., 2018) and numerous studies have documented the impacts of $PM_{2.5}$ on human health (Archer-Nicholls et al., 2016; Chowdhury et al., 2019; J. Liu et al., 2016). According to the U.S. Environmental Protection Agency (EPA)'s National Ambient Air Quality Standards, the daily maximum 8-hr average concentration of ground level ozone (O_3) is set at 0.07 parts per million (ppm) (EPA, 2020). Previous studies have identified the source of air pollution and the human health impacts linked to it for different regions, sectors, and industries (Crippa et al., 2019; Huang et al., 2021; Nawaz et al., 2023; Xiong, Huang, & Du, 2022, Xiong, Partha, et al., 2022). According to Nawaz et al. (2023), the dominant sectors for premature deaths in China are the agricultural and industry, together contributing to 52% of premature deaths for $PM_{2.5}$.

While previous research has explored emissions inventories, none of these studies to our knowledge have specifically examined or compared the disparities in air quality and their resulting impacts on human health arising from the utilization of three distinct global anthropogenic emission inventories and identified the associated drivers. Therefore, this study employed a global chemistry-climate model, the Community Atmosphere Model with chemistry version 6.0 (CAM-chem) within the Community Earth System Model version 2.2.0 (CESM2.2) (Danabasoglu et al., 2020) to quantify the effects of global anthropogenic emissions on air quality and human health. We explored the differences between air quality impacts of three commonly used global emissions inventories; CEDS, CAMS, and ECLIPSE (Bran et al., 2022; Huang et al., 2017), and assessed human health impacts

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estimated from these inventories comparing different regions. This study utilized concentration-response functions from the Global Burden of Disease 2019 study following Xiong et al. (2024) and Xiong, Partha, et al. (2022).

2. Research Methods

2.1. Global Anthropogenic Emission Inventories

2.1.1. The Community Emissions Data System (CEDS)

We used the recently updated CEDS data set, CEDS v2021_04_21 (O'Rourke et al., 2021), that contains historical anthropogenic emissions from 1750 to the year 2019 at the horizontal grid resolution of 0.5° latitude by 0.5° longitude, developed at the Pacific Northwest National Laboratory (Hoesly et al., 2018; O'Rourke et al., 2021; Xiong, Partha, et al., 2022). CEDS assesses emissions in 221 different geographic areas, encompassing eight types of fuels and emissions from 55 distinct industry sectors. These industry sectors within CEDS follow the energy statistics guidelines established by the International Energy Agency. CEDS employs a "mosaic" scaling approach to maintain precise information regarding specific fuels and sectors across different data sets, while ensuring consistent methodologies are applied consistently across various locations and time periods. The use of mosaic inventories has become more common as they allow for the use of detailed local emissions while harmonizing this information across broad regional or global scales (Crippa et al., 2019). Sectors of CEDS emission inventory include agricultural, energy, industrial, transportation, residential, solvents, waste, and international shipping, with the global annual total emissions for various species in 2015 being listed in Tables S1 and S2 in Supporting Information S1.

2.1.2. Copernicus Atmosphere Monitoring Service (CAMS)

We have used the most recently updated CAMS data set, CAMS version 5.1 (CAMSv5.1)'s Global Anthropogenic emissions (CAMS-GLOB-ANT) (Soulie et al., 2024), a combination of CEDS and EDGAR version 5 (EDGARv5). Both CAMS and EDGAR provide emissions on monthly and annual averages with a resolution of $0.1° \times 0.1°$ (Kuenen et al., 2022; Soulie et al., 2022). CAMS provides anthropogenic emissions of main atmospheric pollutants and greenhouse gases that are relevant in our study such as NO_x, SO_2, BC, NH_3 and NMVOCs and provides emissions for 2000–2022 period. For the NMVOCs emissions, each individual species' emission was calculated separately to find a summation of NMVOCs emitted in the year 2015 (Table S3 in Supporting Information S1). In this paper, we refer to CAMSv5.1's CAMS-GLOB-ANT as CAMS. EDGAR methodology for emissions is described in Crippa et al. (2018)and can be found online publicly (https://edgar.jrc.ec.europa.eu/dataset_ap50). CAMS emission inventory has been frequently used in CESM model experiments (Bouarar et al., 2021; Gaubert et al., 2021; Tang et al., 2022, 2023) and in health studies (Bui et al., 2023; Roberts & Wooster, 2021).

2.1.3. Evaluating the Climate and Air Quality Impacts of Short-Lived Pollutants (ECLIPSE)

We also use global anthropogenic emissions using the International Institute for Applied System Analysis Greenhouse Gas-Air Pollution Interactions and Synergies (GAINS) (Amann et al., 2011) integrated assessment model, the most recently updated ECLIPSE (Evaluating the Climate and Air Quality Impacts of Short-Lived Pollutants version 6b) for 2015. All major air pollutants and greenhouse gases from Kyoto Protocol are included in GAINS, with emissions estimates provided for 200 country-regions (Klimont et al., 2017; Stohl et al., 2015). The species in ECLIPSE, which includes the eight specified pollutants we are focusing on, are at spatial resolutions of 0.5° latitude by 0.5° longitude and is re-gridded to match the model spatial resolution (Huang et al., 2018). This data set, which includes international shipping, was updated in February 2021. The temporal distribution ranges from 1990 to 2030 in five-year intervals, 2040, and 2050. This emission inventory includes nine sectors: energy, industry, solvent use, transport, residential combustion, agriculture, open burning of agricultural waste, waste treatment, gas flaring and venting, and international shipping (Table S4 in Supporting Information S1) (Klimont et al., 2025). Previous studies have used ECLIPSE for air quality studies (Huang et al., 2018; Lund et al., 2023; Stohl et al., 2015; Upadhyay et al., 2020) and to estimate the premature deaths due to exposure to $PM_{2.5}$ and O_3 (Huang et al., 2021).

ECLIPSE provides policy-driven emission projections based on anticipated regulatory interventions and sectoral activity but does not dynamically adjust for meteorological feedback (Granier, et al., 2019; Klimont et al., 2017;

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Table 1
Model Simulation Configuration for Cases 1–4

Cases	Anthropogenic emission inventory
Case 1	CEDS
Case 2	CAMS
Case 3	ECLIPSE
Case 4	OFF

Stohl et al., 2015). However, CAM6-Chem can simulate how meteorological conditions influence aerosol formation and chemical transformations (Emmons et al., 2020; Tilmes et al., 2019). ECLIPSE emissions estimates indicate that reductions in sulfate oxides (SO_x) emissions can lead to increased ammonium nitrate (NH_4NO_3) formation, contributing to higher secondary inorganic aerosol concentrations. This effect is particularly evident in ammonia-rich regions like South Asia, where excess NH_3 shifts toward nitrate aerosol production as sulfate levels decline, altering the chemical composition of atmospheric aerosols (Ge et al., 2023).

In CAM-chem, secondary organic aerosol (SOA) formation follows a Volatility Basis Set approach, where biogenic and anthropogenic VOCs oxidize via OH, O_3, and NO_3 radicals, forming semi-volatile species that partition onto aerosols (Emmons et al., 2020; Tilmes et al., 2019). The variability in VOC emissions across different inventories affects SOA production, particularly in regions where biomass burning and anthropogenic emissions dominate such as Africa and Asia (Granier, et al., 2019). ECLIPSE projects higher future NH_3 and NO_x emissions in certain regions, particularly in agricultural regions in Southern Asia and Africa, this could lead to enhanced nitrate aerosol formations where SO_4 levels decline (Klimont et al., 2017).

2.2. CESM2 CAM-Chem Model Simulations

In this study, we used the NSF NCAR (National Center for Atmospheric Research) Community Atmosphere Model version 6.0 with chemistry (CAM-chem). CAM-chem is a configuration of CESM2 (Gettelman et al., 2019) and in this study we used CESM2 version 2.2.0. CAM-chem is coupled with the Community Land Model (CLM) version 5.0 (Iacono et al., 2008; Lawrence et al., 2019). CAM-chem uses Model for Ozone and Related chemical Tracers as gas-phase chemical mechanisms to represent the detailed oxidation and reaction pathways for reactive species in the troposphere and stratosphere (Emmons et al., 2020). For aerosol micro-physical processes, CAM-chem uses modal aerosol model with 4 modes (MAM4), which includes Aitken, accumulation, primary carbon and coarse modes (Jo et al., 2023; X. Liu et al., 2016; Tilmes et al., 2019). The aerosol size distributions in each mode of MAM4 are assumed to be lognormal (X. Liu et al., 2016).

The CLM simulates mineral dust emissions using a physically based parameterization driven by surface wind stress, soil properties, and land cover (Leung et al., 2024; Zender et al., 2003). Dust emission fluxes from CLM are then passed to CAM-chem to account for transport, deposition, and microphysics (X. Liu et al., 2016).

To quantify the impacts of global anthropogenic emissions on air quality modeling, we ran four CAM-chem simulations from 1 January 2014, to 1 January 2016, at the horizontal resolution of 0.95° latitude by 1.25° longitude, using different anthropogenic emission inventories (Table 1; Text S1 in Supporting Information S1). In our model simulations we use historic global biomass burning emissions for CMIP6 (BB4CMIP) for fire emissions (Van Marle et al., 2017). Biogenic emissions are internally derived using the Model of Emissions of Gases and Aerosols from Nature (Emmons et al., 2020; Guenther et al., 2012; Jo, Tilmes, et al., 2023). CAM-chem is nudged every 12 hr to the Modern-Era Retrospective analysis for Research and Applications version 2 meteorological data set. We used the first year of the simulation as spin-up and the rest for result analysis.

In Table 1, Cases 1–3 shown represent the simulations with anthropogenic emission inventory from CEDS, CAMS and ECLIPSE, respectively. To quantify the net effect of global anthropogenic emissions, we ran the Case 4, in which we turned off the global anthropogenic emissions, with other model configurations being the same as Cases 1–3. The differences of Cases 1–3 and Case 4 will be the net effects from anthropogenic emissions for CEDS, CAMS and ECLIPSE, respectively.

Before computing mortality rates, the CAM-chem model's outputs for $PM_{2.5}$ and O_3 concentrations are down-scaled from the original resolution of 0.9° latitude by 1.25° longitude to a finer resolution of 0.1° latitude by 0.1° longitude. This aligns the outputs with the horizontal and gridded resolution of both the total population counts and the country mask. The CAM-chem simulated $PM_{2.5}$ concentrations underwent downscaling using a recently released satellite-derived $PM_{2.5}$ product. For CAM-chem $PM_{2.5}$ outputs, we have downscaled using the newly available satellite data that is at 0.1° latitude x 0.1° longitude resolution. We minimized the negative biases between modeled and observed $PM_{2.5}$ by adjusting the $PM_{2.5}$ concentrations (Van Donkelaar et al., 2021).

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2.3. Calculations of Human Health Effects: $PM_{2.5}$ and O_3 Induced Premature Deaths

We used the Global Exposure Mortality Model (GEMM) (Burnett et al., 2018), as described in Huang et al. (2021), to calculate the regional and global annual premature deaths attributable to long-term ambient surface $PM_{2.5}$ exposure. GEMM predicted non-accidental deaths from lower respiratory infections (LRIs) and noncommunicable diseases (NCDs) linked to long term exposure to $PM_{2.5}$.The calculation of the Hazard Ratio (HR) function in GEMM is performed for each specific combination of longitude (i), latitude (j), health endpoint (h), and 5-year age interval group (a) at every horizontal model grid box is calculated as expressed below in Equation 1 (Huang et al., 2021).

$$HR_{i,j,h,a} = \exp\left\{\frac{\theta\, ln\left(\frac{z_{i,j}}{\alpha}+1\right)}{1+\exp^{\left(\frac{z_{i,j}-\mu}{\upsilon}\right)}}\right\}, z_{i,j} = \max(0, C_{i,j} - C_0) \quad (1)$$

θ, μ, α and υ are the GEMM model fit parameters which are obtained from Burnett et al. (2018). We use the standard errors θ to estimate the uncertainty and 95% confidence interval (CI) of $PM_{2.5}$-induced premature deaths. $C_{i,j}$ represents the average yearly surface concentration of $PM_{2.5}$ in a specific grid cell denoted by (i, j). C_0 is the hypothetical $PM_{2.5}$ concentration set at 2.4 μg/m^3, below which it is assumed that there is no risk of premature death.

For O_3-induced mortality, we followed studies by Huang et al. (2020) and Turner et al. (2016) to calculate the O_3-mortality HR function associated with chronic obstructive pulmonary disease (COPD). The study conducted by Turner et al. (2016) links between long-term exposure to O_3 and an increased risk for respiratory mortality, and it is found that for every 10 ppb increase in O_3 levels the respiratory mortality risk is increased by 12%. The O_3-mortality HR function in each grid cell (i, j) for each health endpoint (h) and each age group (a) is then calculated as:

$$\mathrm{HR}_{i,j,h,a} = \exp^{\eta \Delta Y_{i,j}}, \Delta Y_{i,j} = \max(0, \Delta Y_{i,j} - 26.7) \quad (2)$$

where η represents the log-linear slope between O_3 concentrations and the HR associated with COPD, with a mean value of 0.0131 (95% CI: 0.0077–0.0191). $Y_{i,j}$ denotes the annual mean Maximum Daily 8-hr Average (MDA8) O_3 concentration in grid cell (i, j). It is assumed that the threshold value of MDA8 O_3 concentration is 26.7 ppb, below which there is no mortality risk associated with O_3-induced COPD (Huang et al., 2021; Turner et al., 2016).

The CAM-chem model simulation outputs are regridded to 0.1° latitude and 0.1° longitude from their 0.9° latitude x 1.25° longitude resolution, which is required for the country mask and gridded total population count. To assess impact on human health caused by exposure to ambient $PM_{2.5}$ and O_3 we consider the baseline mortality rate (BMR) from the latest Global Burden of Disease 2019 (hereafter referred to GBD 2019) (GBD 2019 Risk Factors Collaborators, 2020), the gridded population density (POP), and the HR as shown in Equation 3:

$$M_{i,j,h,a} = \mathrm{POP}_{i,j,a} \times \mathrm{BMR}_{i,j,h,a} \times \left[\left(\mathrm{HR}_{i,j,h,a} - 1\right)/\mathrm{HR}_{i,j,h,a}\right] \quad (3)$$

where $M_{i,j,h,a}$ is the gridded annual total premature deaths (APDs) associated with $PM_{2.5}$ and O_3 exposure at each grid box (i, j) for each health endpoint (h) and age group (a); $\mathrm{POP}_{i,j,a}$ denotes gridded POP which is estimated as the product of population counts in each grid cell (i, j) and the age specific (5 years interval) population fraction.

Following Huang et al. (2021), the gridded POP is sourced from the Gridded Population of the World version 4 for the year 2015 (Center for International Earth Science Information Network – CIESIN – Columbia University, 2018) at a horizontal resolution of 2.5 min × 2.5 min. This data is then regridded to a resolution of 0.1° latitude × 0.1° longitude.

The Years of Lives Lost (YLL), for each grid cell (i, j), health endpoint (h) and age group (a) are calculated in Equation 4 as:

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Table 2
Global Annual Total Emissions of Various Species for the Year 2015 From CEDS, CAMS and ECLIPSE

	BC	OC	NO_2	CO	NMVOC	NH_3	SO_2
CEDS	6.0	14.0	121.9	565.6	149.6	59.4	92.1
CAMS	4. 8	16.5	121.5	578.1	148.3	49.1	99.7
ECLIPSE	6.4	13.9	125.0	548.4	112.0	61.0	73.4

Note. Units: Tg specie per year. In CAMS inventory, NO_x was archived as NO, and we converted it to NO_2 for comparing with CEDS and ECLIPSE.

$$YLL_{i,j,h,a} = M_{i,j,h,a} \times MYLL_{i,j,h,a} \quad (4)$$

$MYLL_{i,j,h,a}$ is the mean YLL for each health point and each age group attributed to all causes from GBD 2019 for the year 2015.

Following previous studies (Huang et al., 2021; Xiong, Partha, et al., 2022), we divide the globe into 11 regions based on the classification by the International Monetary Fund (IMF): China, India, United States of America (USA), Canada, Eastern and Central Europe (ECEUROPE), Western Europe (WEUROPE), Latin America (LATIN), North Africa and Middle East (NAME), Rest of Asia (ROA), Sub-Saharan Africa (SSA) and Rest of the world (ROW). We calculated the regional and global annual health burdens of exposure to $PM_{2.5}$ and O_3 by adding up each grid cell for each simulation (Huang et al., 2020, 2021). We subtracted all total mortalities in each simulation from Cases 1–3 from Case 4 to quantify the net health effects for each global anthropogenic emission inventory (CEDS, CAMS and ECLIPSE).

3. Results

3.1. Intercomparisons of Regional and Global Spatial Patterns of Emissions Among CEDS, CAMS and ECLIPSE for the Year 2015

Table 2 lists the annual total emissions of BC, OC, NO_2, CO, NMVOC, NH_3, and SO_2, respectively, from CEDS, CAMS and ECLIPSE for the year 2015. The total emissions from the CAMS and ECLIPSE inventories were then compared to the total emissions using CEDS as baseline. In general, global annual total emissions of NO_2 and CO for the year 2015 are consistent among CEDS, CAMS and ECLIPSE (Table 2).

In terms of O_3 precursors such as CO, we discovered that CO accounted for 56% (CEDS), 57% (CAMS), and 58% (ECLIPSE) of the respective global annual total anthropogenic emissions (Figure 1) in 2015 (Figure 1d). The annual total emissions of NMVOC between CEDS (149.6 Tg) and CAMS (148.3 Tg) were also consistent, with values from ECLIPSE for NMVOC 25% lower than that from CEDS (Figure 1e) (Placet et al., 2000). Large discrepancies associated with global annual total emissions of NH_3 in 2015 between CEDS and CAMS were found, with the latter 20% lower than the former (Figure 1f). In terms of $PM_{2.5}$ precursors we find that globally the annual total of BC emissions were 6.0 (CEDS), 4.8 (CAMS) and 6.4 Tg/yr (ECLIPSE), while OC was 13.9 (CEDS), 16.5 (CAMS) and 13.8 Tg/yr (ECLIPSE) (Contini et al., 2018). Compared with CEDS, the global annual total emissions of BC from CAMS and ECLIPSE are 21% lower and 6% higher, respectively (Figure 1a). For OC, the global annual total emissions between CEDS (14.0 Tg) and ECLIPSE (13.9 Tg) in 2015 were comparable. However, we found that the global annual total OC emissions for the year 2015 from CAMS were about 20% higher than that from CEDS (Figure 1b). Nevertheless, NH_3 emissions from CEDS and ECLIPSE were consistent. Global annual total emissions of SO_2 from CAMS and ECLIPSE for the year 2015 were 8% higher, and 20% lower than that from CEDS, respectively (Figure 1g).

Previously, Hoesly et al. (2018) found that CEDS estimates were generally larger than those of EDGARv4.3 for global estimates across most species. In our comparison, we found that the global annual total emissions of BC and NH_3 from CEDS (Tables S1 and S2 in Supporting Information S1) in 2015 are much higher than those from CAMS (Table S3 in Supporting Information S1), which could be attributable to differences in sector distributions (Hoesly et al., 2018; Kuenen et al., 2022).

3.2. Validations of Model Simulations Against Observations

To validate our model simulation, we employed surface $PM_{2.5}$ measurement networks from the US, China, India, and Europe. Subsequently, we compared these measurements with the results of our CAM-chem model simulations for the year 2015. Following the methodology of Huang et al. (2021) and Xiong, Partha, et al. (2022), we calculated the annual average surface $PM_{2.5}$ concentrations. Measurements from the US, Europe, India, and China were used in these assessments. We specifically obtained detailed $PM_{2.5}$ annual concentration data for the United States (US) from the Improved Air Quality System network, downloaded from the EPA's Air Quality System (AQS) Data Mart (https://aqs.epa.gov/aqsweb/airdata/download_files.html). Annual $PM_{2.5}$ concentrations from India were collected from the National Ambient Air Quality Monitoring Program (https://cpcb.nic.in/namp-data/).

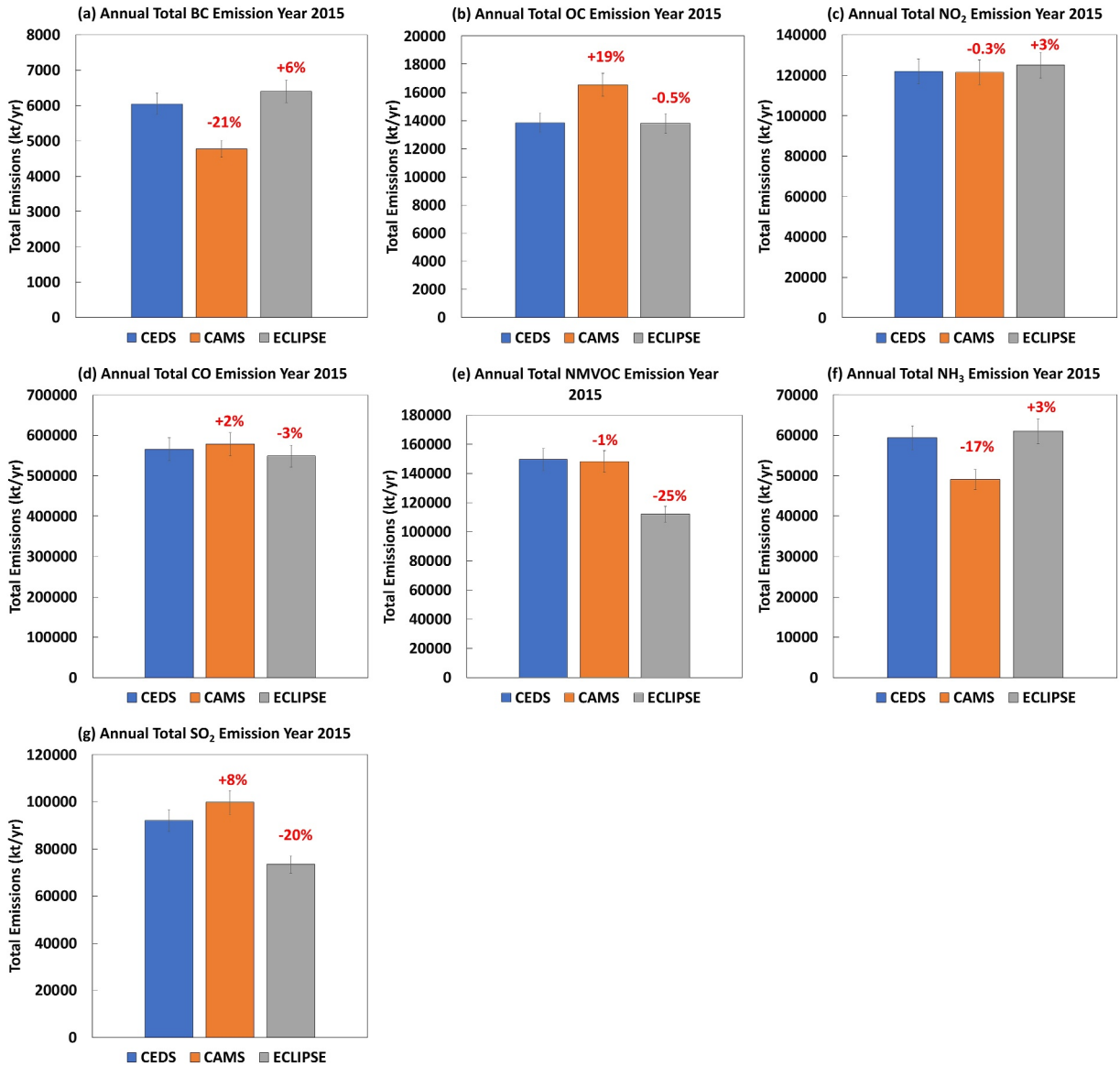


Figure 1. An inter-comparison of global annual total emissions of (a) black carbon, (b) organic carbon, (c) NO_2, (d) CO, (e) non-methane volatile organic compound, (f) NH_3, and (g) SO_2 for the year 2015 from Community Emissions Data System (blue), Copernicus Atmosphere Monitoring Service (orange) and Evaluating of the Climate and Air Quality Impacts of Short-Lived Pollutants (gray). Units: kt specie/yr.

We have also used annual $PM_{2.5}$ concentration data from the European Environment Agency's Air Quality e-Reporting system for Europe (EEA; https://www.eea.europa.eu/en/analysis/maps-and-charts/air-quality-statistics-dashboards). A total of 142, 224, 56, and 74 ground stations were chosen in US (Figure 2a), Europe (Figure 2b), India (Figure 2c), and China (Figure 2d), respectively.

To compare observed data and the model's $PM_{2.5}$ simulation in the four different regions, we used statistical techniques such as normalized mean bias (NMB) and Pearson's correlation coefficient (r).

When validated against ground observational data, the NMB values calculated for CEDS, CAMS, and ECLIPSE reveal distinct patterns across various regions. Across Europe, all three emission inventories consistently underestimate surface $PM_{2.5}$ concentrations with NMB values of −43.3% (CEDS), −43% (CAMS), and −50% (ECLIPSE). Similar underestimations are observed in India (CEDS: −44.1%, CAMS: −41.9%, ECLIPSE: −51.0%) and China (CEDS: −66.1%, CAMS: −53.5%, ECLIPSE: −75.3%). In contrast, the models perform better over the US, with NMB values of −7.4% (CEDS), +1.6% (CAMS), and −13.5% (ECLIPSE), indicating

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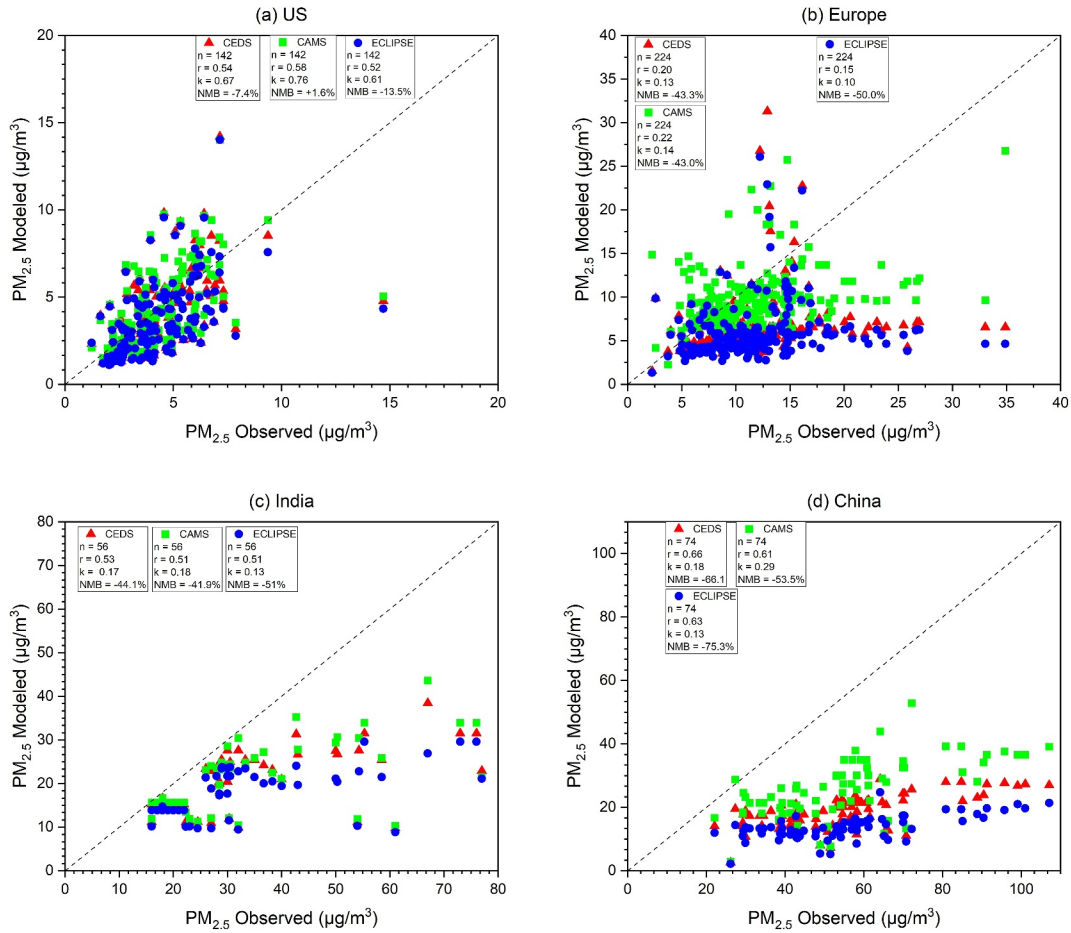


Figure 2. Scatter plot of observed and model surface $PM_{2.5}$ concentrations; Community Emissions Data System (red triangle), Copernicus Atmosphere Monitoring Service (green square), and Evaluating of the Climate and Air Quality Impacts of Short-Lived Pollutants (blue circle) values over (a) United States, (b) Europe, (c) India, and (d) China in 2015. The dashed line is 1:1 reference line.

relatively minor biases. ECLIPSE exhibits the largest negative biases across Europe, India, and China, although in the US, it shows a moderate underestimation compared to CEDS and CAMS. Among the three inventories, CAMS demonstrates the least bias overall, with smaller under- and overestimations across all regions.

CESM2 underestimates surface mineral dust emissions over source regions (Leung et al., 2025). This underestimation is especially pronounced in regions such as East Asia, where observed dust levels are significantly higher than those simulated (Wu et al., 2019). This uncertainty may partially contribute to the underestimates of model simulated surface $PM_{2.5}$ concentrations, compared with observations.

In an earlier version of the CAM-chem model simulation, CAM5-Chem using the ECLIPSE V5a inventory in Huang et al. (2021) observed that surface $PM_{2.5}$ concentrations attributed to higher SOA levels compared to observations (Tsigaridis et al., 2014). Similar overestimations persist in CAM6-chem as identified by J. Liu et al. (2020) and Y. Liu et al. (2020). Another likely reason for overestimation can also be due to the summertime overestimates of sulfate concentrations (Tilmes et al., 2015), thereby compounding the overall $PM_{2.5}$ overestimation. Consistent with these findings, Xiong, Partha, et al. (2022) reported NMB values of −57.7%, −67.7%, and −44.4% for CEDS over Europe, China and India, respectively.

As shown in Section 3.1 (Figures 1c, 1e, and 1g), precursors linked with formation of $PM_{2.5}$, such as NO_2, NMVOCs, NH_3, and SO_2, are much lower in ECLIPSE (except for NO_2) compared to CEDS and CAMS. Consistently, the relative difference of NMB for CEDS when compared to ECLIPSE is −81% lower, while for CAMS the NMB relative difference is +78% higher. The differences in the different emission inventories and

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their impact on aerosol formation critically influence the representation of aerosols in climate models (Jo et al., 2021).

To verify the accuracy of CAM-chem simulated O_3 concentration, we employed the surface O_3 concentrations from the Tropospheric Ozone Assessment Report (TOAR) II (Huang et al., 2021; Xiong, Partha, et al., 2022). This involved regridding the model from 0.9° × 1.25° longitude-latitude grid to 2° × 2° latitude-longitude grid and comparing it with TOAR. We have used an updated TOAR data, TOAR II, which includes 2015 (Fiore et al., 2009; Schultz et al., 2017; Young et al., 2013, 2018), allowing for more direct comparisons. The generated data set focused on rural observational data and focused on surface O_3 observations. There were notable discrepancies between the surface O_3 concentrations predicted by the model simulations and the actual observed values reported in the TOAR data, as illustrated in Figure 3. We found that the model overestimated surface O_3 concentrations compared with TOAR observations, with mean biases of 6.6 ppbv for CEDS (Figure 3a), 5.6 ppbv for CAMS (Figure 3b), and 2.4 ppbv for ECLIPSE (Figure 3c), respectively. The *r* coefficients were 0.67, 0.67, and 0.66 for CEDS, CAMS and ECLIPSE respectively. Previous studies have reported similar overestimations (Huang et al., 2021; Lamarque et al., 2012; Tilmes et al., 2015).

It is important to note that the Huang et al. (2021) study used an earlier version of the ECLIPSE emissions inventory (ECLIPSEv5a), whereas our study utilizes the updated inventory (ECLIPSEv6b). As a result, our reported mean bias (2.4 ppbv) is substantially lower than that reported by Huang et al. (2021), which was 17.9 ppbv. One of the reasons for the differences with the updated ECLIPSE emissions inventory could be due to the revised international shipping, updates for several sectors such as power plants, flaring, transportation industry-all of which are sources for NMVOCs and NO_x, which contribute to the formation of O_3. Another reason is that there was an update on the general legislation and historical data especially for the year 2015 (https://iiasa.ac.at/models-tools-data/global-emission-fields-of-air-pollutants-and-ghgs).

We recognize that our simulations using CEDS and CAMS emissions inventories exhibit high biases in surface O_3 concentrations. Consequently, this would impact the estimated premature deaths due to anthropogenic activities associated with chronic obstructive pulmonary disease (COPD) attributable to long-term surface O_3 exposure. Our simulation using ECLIPSE emissions inventory demonstrated a significantly lower mean bias when compared to a previous study (Huang et al., 2021) indicating an improvement. Lastly, it's important to highlight that many regions lack recorded observational data, resulting in unfilled or blank grid cells on the maps.

3.3. Air Quality Impacts

In the following section, we examined the impact of the three anthropogenic emissions inventories on global ambient surface concentrations of O_3, NO_x, and $PM_{2.5}$ respectively for the year 2015, while subtracting the Case 4 from each of the simulations with different emission inventories. Other details, including changes in BC, CO, NMVOC, NH_3 and SO_2 are provided in Text S2 in Supporting Information S1.

Overall, a majority of CEDS emission sources come from Asia, Europe, and North America. Similar spatial patterns are found for ECLIPSE and CAMS (Figures 4–6 and Figures S2–S6 in Supporting Information S1). However, the spatial differences in pollutants such as O_3 and $PM_{2.5}$ are more substantial between CEDS and ECLIPSE than between CEDS and CAMS.

Figures 4–6 show the global spatial distribution of annual average concentration of O_3, NO_x and $PM_{2.5}$ for the year 2015 (see Figures S2–S6 in Supporting Information S1 for BC, CO, NMVOC, NH_3 and SO_2 species).

NMVOCs and NO_x are the primary precursors leading to net O_3 production (McDuffie et al., 2020). The global annual mean surface O_3 concentrations driven by anthropogenic activities are 9.3, 8.1, and 6.7 ppbv for CEDS, CAMS and ECLIPSE respectively. For surface O_3 impacts, we found that annual surface O_3 concentrations are highest over Asia, Northern Africa, and the western coast of North America (Figure 4). The global spatial variability of surface O_3 concentrations is primarily driven by meteorological conditions rather than the spatial distribution of its precursors, such as NO_x, due to its longer atmospheric lifetime (Ansari et al., 2016). When examining the distribution of NO_x across all three inventories (Figure 5), the patterns appear comparable, with global mean concentrations of 0.4, 0.4, and 0.5 ppbv for CEDS, CAMS, and ECLIPSE respectively. A higher NO_x concentration was consistently observed in Eastern Asia across three inventories.

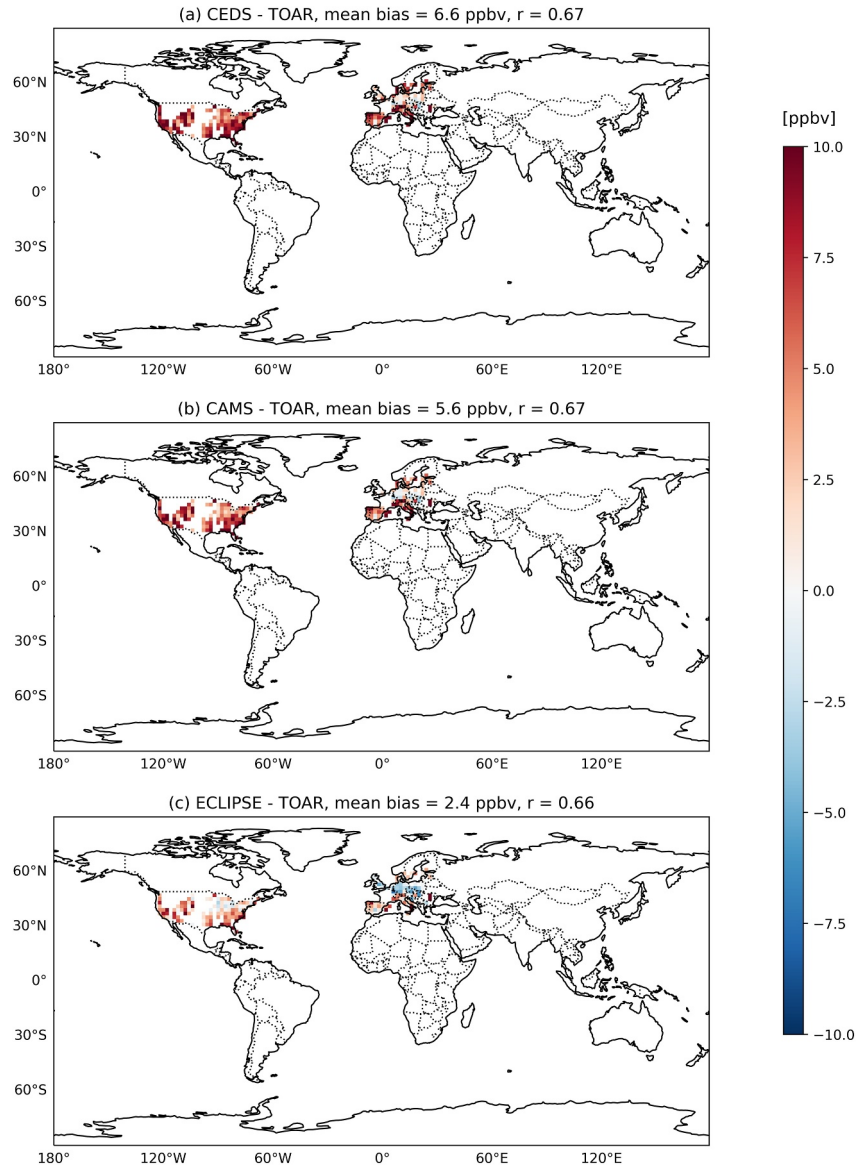


Figure 3. The comparison between model simulations of surface O_3 concentrations using the Community Emissions Data System, Copernicus Atmosphere Monitoring Service and Evaluating of the Climate and Air Quality Impacts of Short-Lived Pollutants emissions inventories and Tropospheric Ozone Assessment Report with (a) CEDS-TOAR, (b) CAMS-TOAR, and (c) ECLIPSE-TOAR. The average mean bias is determined as the mean variation in each grid cell when both observations and simulations are available for the year of 2015. Unfilled/White grid cells indicate areas without recorded measurements.

The three simulations revealed a similar distribution of surface $PM_{2.5}$, with higher concentrations in Asia, Africa, South America, North America, and Australia (Figure 6).We found that annual mean surface $PM_{2.5}$ concentrations, driven by anthropogenic activities, exhibited substantial burdens over India at 13.8 (CEDS), 14.2 (CAMS), and 11.1 (ECLIPSE) $\mu g/m^3$, and over China at 5.5 (CEDS), 8.0 (CAMS), and 3.9 (ECLIPSE) $\mu g/m^3$ (Table S5 in Supporting Information S1). India contributes on average 43% of total $PM_{2.5}$ concentrations attributable to anthropogenic activities across the three inventories. Large surface impacts of BC impacts are observed over South Asia, Eastern Asia, Africa, and South America (Figure S2 in Supporting Information S1), which are consistent with previous studies (Huang et al., 2018, 2020, 2021). For CO (Figure S3 in Supporting Information S1), we find that the annual global mean surface concentration for CEDS, CAMS, and ECLIPSE majority of the impacts are found in Eastern Asia. The spatial distributions of NMVOCs (Figure S4 in Supporting Information S1) and NH_3 (Figure S5 in Supporting Information S1) are similar across all inventories, with elevated

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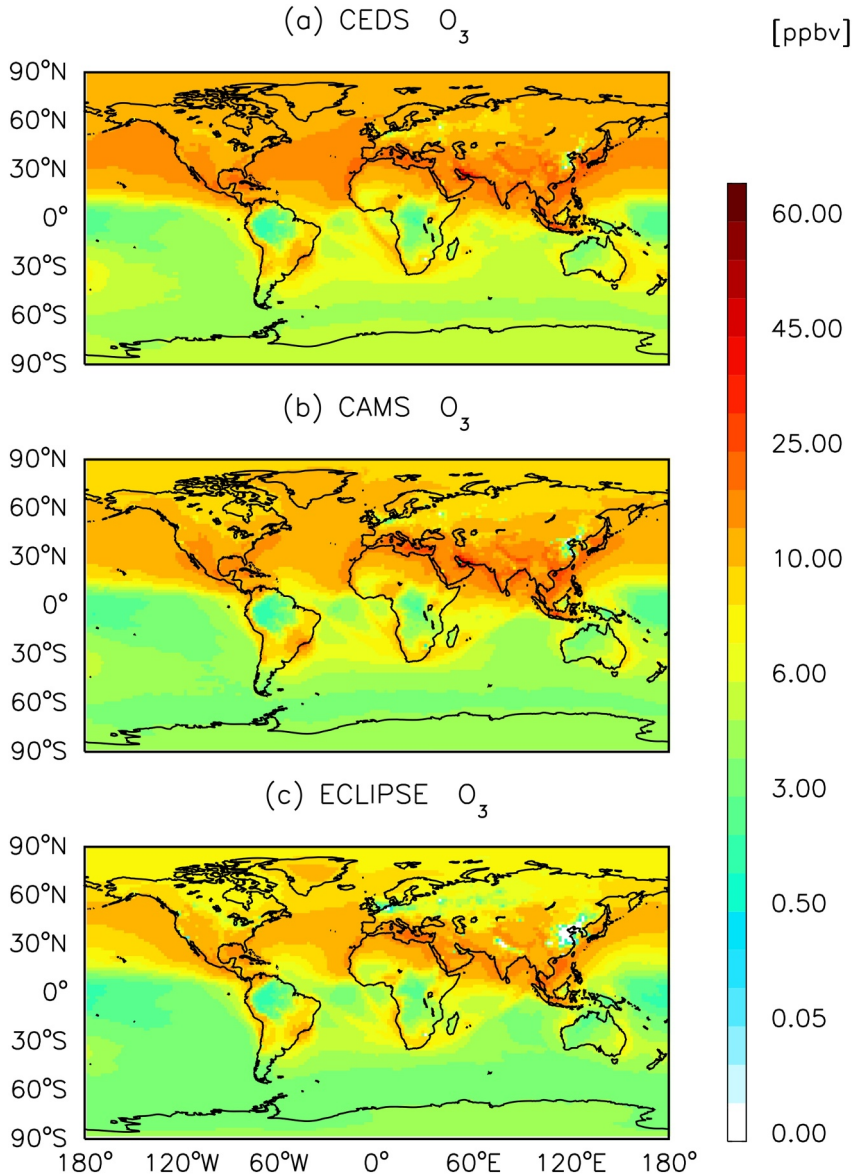


Figure 4. Global annual mean surface O_3 concentrations for the year 2015 from CESM2.2 CAM-chem, with anthropogenic emission inventories of (a) CEDS, (b) CAMS, and (c) ECLIPSE.

concentrations in South America, Africa, and Eastern Asia. ECLIPSE's simulated concentrations for NMVOCs are lower than CAMS and CEDS, averaging 3.1 ppbv annually. In all three simulations, SO_2 (Figure S6 in Supporting Information S1) concentrations are predominantly over Asia (Eastern and Southern), with CAMS showing higher impacts in Eastern Asia. The differences in spatial distributions of global annual mean surface $PM_{2.5}$ and O_3 can be found in Figures S7 and S8 in Supporting Information S1.

3.4. Impacts on Human Health

Global APDs following Huang et al. (2021) methodology is provided in Table 3. The estimated global $PM_{2.5}$- and O_3-induced APDs attributed to anthropogenic activities for CEDS, CAMS and ECLIPSE are 2.5 (95% CI: 2.1–3.1), 2.8 (95% CI: 2.4–3.4), and 1.9 (95% CI: 1.6–2.4) million (Figure 7). Regionally, $PM_{2.5}$- and O_3-induced mortalities from anthropogenic activities peak in China, India, and ROA (Figure 8). On average, China represents 33.8%, 38.9%, and 30.0% of the global total premature mortality in CEDS, CAMS, and ECLIPSE, respectively.

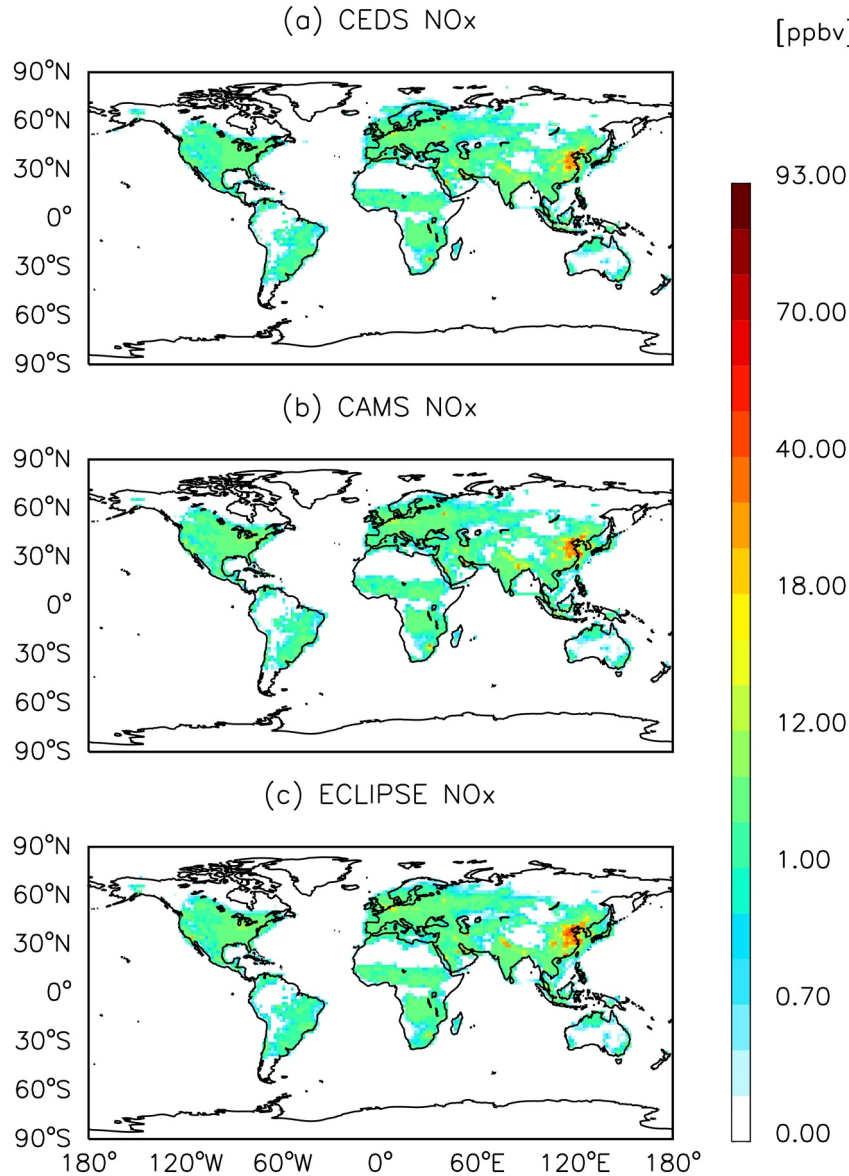


Figure 5. Same as Figure 4 but for NO_x.

For India, these figures are 24.9%, 23.2%, and 26.4% across the three inventories, while ROA accounts for 15.4%, 14.1%, and 15.2% in CEDS, CAMS, and ECLIPSE, respectively.

When comparing between the inventories' $PM_{2.5}$-induced APDs, we use CEDS as the baseline for comparisons. Regionally we find that China and US have the highest percentage discrepancy compared to CAMS, with an average difference of +41.5% for China and +22.8% for US (Table S6 in Supporting Information S1). In CAMS, we find that the NAME region has the lowest $PM_{2.5}$ APDs percentage difference at −10.8%. ECLIPSE generally showed lower percentage differences, with notable values in regions like China (−23.9%), Eastern Europe (−15.9%), Latin America (−28.0%), and Western Europe (−23.5%). However, SSA showed a positive percentage difference (+13.6%) for ECLIPSE.

Regionally, when comparing the O_3-induced deaths from the three inventories, we observe a discernible pattern for ECLIPSE, as the percentage differences for O_3 APDs (Table S7 in Supporting Information S1) are significantly larger compared to the baseline, with China showing the highest difference at −57.0% and the ROW region coming in second at −48.1%. ECLIPSE has the lowest overall total O_3 APDs across all regions when compared to the

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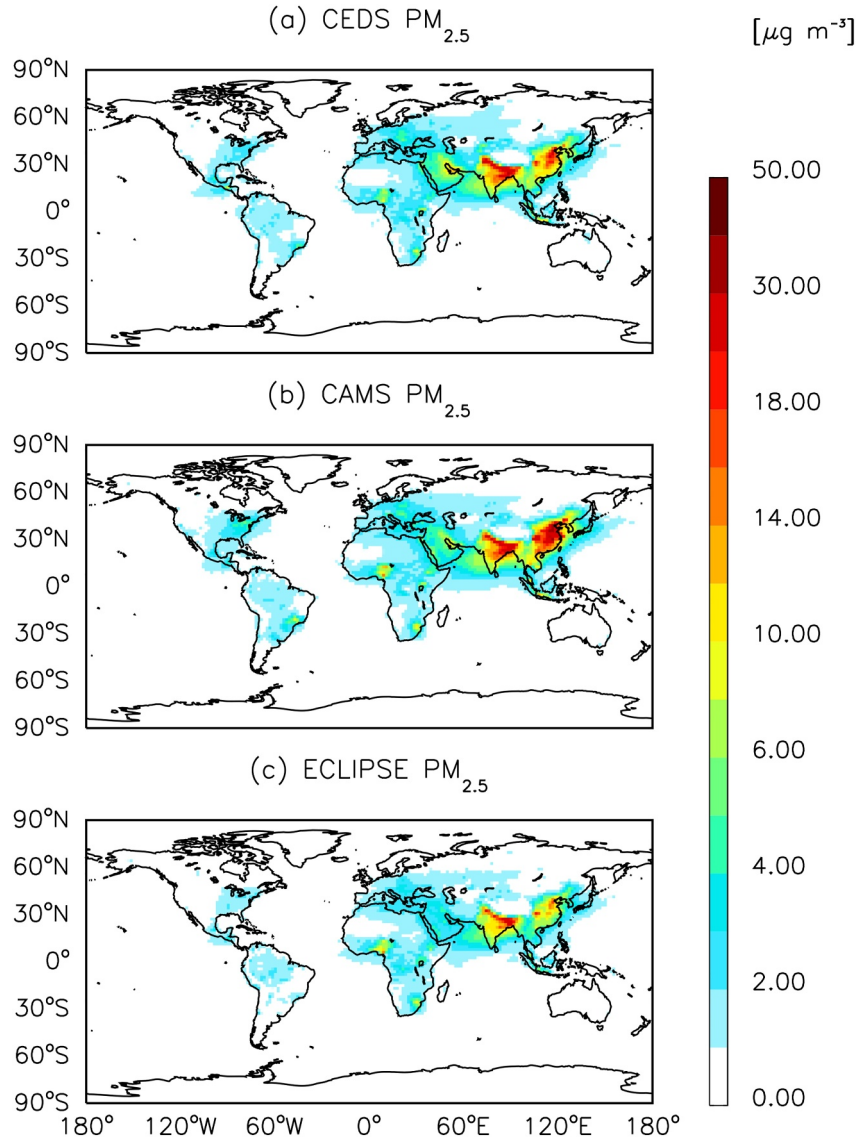


Figure 6. Same as Figure 4 but for $PM_{2.5}$.

baseline. For CAMS, the percentage difference for all regions is less than 10%. Both ECLIPSE and CAMS show a significant percentage difference in the ROW region, at −29% and −6%, respectively. Please refer to Tables S6 and S7 in Supporting Information S1 for detailed APD differences between the three emissions inventories.

YLL associated with $PM_{2.5}$ + O_3 in 2015 were calculated to be 55.4 (95% CI: 50.6–64.1), 62.7 (95% CI: 54.5–70.6) and 62.1 (95% CI: 45.2–85.4) million years CEDS, CAMS, and ECLIPSE, respectively. The YLLs (Table 3) for CAMS are significantly higher than the baseline, and this trend is also seen in the APDs. On a regional scale we find that China, India an ROA have the largest $PM_{2.5}$ mortalities in 2015, similarly the case for O_3-induced mortalities. However, we notice an increase in the mortalities for $PM_{2.5}$ for the CAMS inventory, with significance in China, USA, and Canada at +42%, +23% and +12% percentage difference. Consistently, ECLIPSE has the largest percentage difference across all regions, with a much lower global annual $PM_{2.5}$-induced deaths-the largest difference being in LATIN region at −28% followed by China at −24% difference.

Following Xiong, Partha, et al. (2022), we investigated the impact the economic disparities would have on mortality due to $PM_{2.5}$ and O_3. We had differentiated our data between developing and developed nations using data from the IMF. In this study we have found that developing nations have a higher $PM_{2.5}$- and O_3-induced

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Table 3
Global Annual Total $PM_{2.5}$- and O_3-Induced Premature Deaths and Years of Life Lost in 2015

Cases	Species	2015	
		Premature deaths (x1,000 persons)	Years of lives lost (10^6 years)
CEDS	O_3	695.2 (445.7, 948.0)	11.7 (6.6, 15.6)
	$PM_{2.5}$	1852.0 (1698.1, 2142.7)	43.6 (41.8, 48.4)
	$PM_{2.5} + O_3$	2547.3 (2142.7, 3115.3)	55.4 (50.6, 64.1)
CAMS	O_3	670.6 (429.1, 914.7)	11.4 (7.3, 15.1)
	$PM_{2.5}$	2140.7 (1964.7, 2500.5)	51.4 (47.2, 55.4)
	$PM_{2.5} + O_3$	2811.3 (2393.8, 3415.2)	62.7 (54.5, 70.6)
ECLIPSE	O_3	422.4 (263.4, 599.8)	25.6 (11.7, 46.0)
	$PM_{2.5}$	1499.7 (1374.0, 1755.2)	36.5 (33.4, 39.4)
	$PM_{2.5} + O_3$	1922.1 (1637.3, 2355.0)	62.1 (45.2, 85.4)

Note. The parenthesis denotes the 95% confidence intervals.

mortalities per capita ($\times 10^5$) than developed nations (Table 4). For 2015, we have estimated that the $PM_{2.5}$- and O_3-induced mortalities global APD rate per capita ($\times 10^5$) is 2.3 (95% CI: 1.9–2.8), 2.5 (95% CI: 2.1–3.0), and 1.9 (95% CI: 1.5–2.2) per 100,000 for CEDS, CAMS and ECLIPSE respectively in developed nations. In developing nations, the global $PM_{2.5} + O_3$ APD rates per 100,000 are 23.1 (95% CI: 19.5–28.5) for CEDS, 25.6 (95% CI: 23.0–31.2) for CAMS, and 17.4 (95% CI: 14.8–21.2) for ECLIPSE.

Our study reveals that in developed nations, the ECLIPSE $PM_{2.5} + O_3$ associated APD rate is, on average, 20.4% lower than that of CEDS, while the CAMS APD rate is 7.2% higher than that of CEDS. We found that the $PM_{2.5} + O_3$ APD rate in developing nations on average is 11 times greater than those in developed nations across the three inventories; at 9.9 (CEDS), 10.2 (CAMS), and 9.3 (ECLIPSE). Additionally, when comparing with CEDS as the baseline, ECLIPSE is, on average, 17.2% lower, while CAMS is 12.9% higher.

3.5. Discussion

We have employed a state-of-the-art chemistry-climate model, CAM-chem to quantify the impacts of anthropogenic emission on surface air quality and the associated premature deaths for the year 2015 based on three commonly used emissions inventories CEDS, CAMS and ECLIPSE.

Figures 9 and 10 provide a visual representation of previous studies in comparison to this study. For 2015, our global estimate of O_3 related mortality due to COPD was 695,000 (95% CI: 445,000–948,000), 671,000 (95% CI: 429,000–915,000), 422,000 (95% CI: 263,000–600,000) deaths for CEDS, CAMS and ECLIPSE, respectively (Figure 9).

The estimations using the CEDS and CAMS emissions inventory agree with the 700,000 deaths reported by Anenberg et al. (2010) associated with O_3 exposure, which utilized EDGAR emissions inventory (Horowitz, 2006), for the year 2000 (Figure 9). Similarly, a study by Lelieveld et al. (2015) also used EDGAR emissions for their 2010 analysis. They reported global annual 142,00 deaths in 2010 which is significantly lower (Figure 9). While a study by Silva et al. (2016), using the Representative Concentration Pathway 8.5 (RCP 8.5) emissions inventory based on EDGAR data, reported 470,000 deaths in 2005 which is closer to our estimate using ECLIPSE emissions inventory (Figure 9).

Furthermore, our estimates for global $PM_{2.5}$-induced premature deaths due to LRI and NCD in 2015 were 1.8 (95% CI: 1.7–2.1), 2.1 (95% CI: 2.0–2.5), 1.5 (95% CI: 1.4–1.8) million for CEDS, CAMS and ECLIPSE, respectively (Figure 9). Our global estimation of $PM_{2.5}$-induced deaths for CEDS and CAMS agrees with Silva et al. (2016), which estimated 2.2 (95% CI: 1.0–3.3) million APDs for 2005. However, study by Anenberg et al. (2010) estimated 3.7 million APDs attributable to anthropogenic $PM_{2.5}$, which is significantly higher than all three of our estimates (Figure 9).

Previous studies have also estimated $PM_{2.5} + O_3$ anthropogenic APDs as 2.7 (Silva et al., 2016), and 3.3 (Lelieveld et al., 2015) million for 2005 and 2010 respectively, which are close to the APDs estimations in this

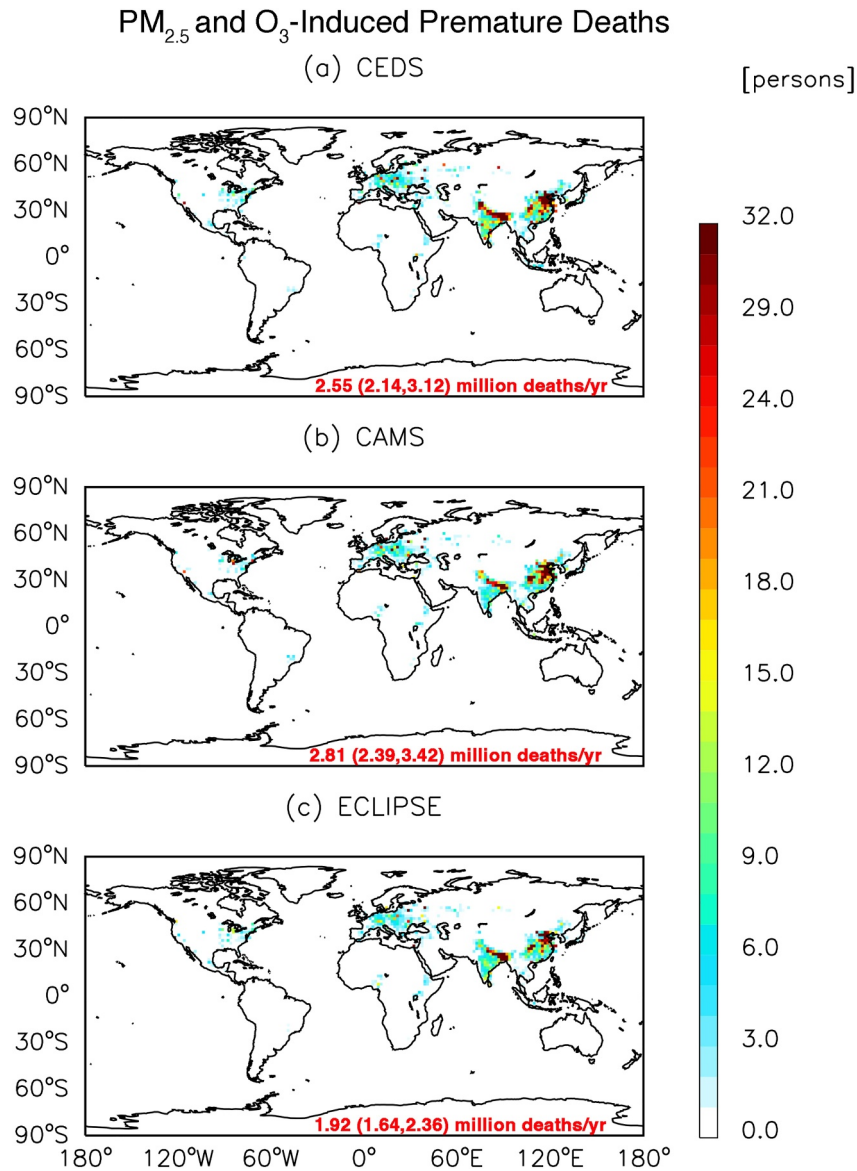


Figure 7. Global annual total $PM_{2.5}$- and O_3-induced premature deaths due to anthropogenic air pollution sources for the year 2015, for (a) Community Emissions Data System, (b) Copernicus Atmosphere Monitoring Service and (c) Evaluating of the Climate and Air Quality Impacts of Short-Lived Pollutants, respectively. Parenthesis shows the uncertainty and 95% confidence intervals.

study at 2.5 (95% CI: 2.1–3.1), 2.8 (95% CI: 2.3–3.4), and 1.9 (95% CI: 1.6–2.3) million for CEDS, CAMS and ECLIPSE, respectively (Figure 9).

Regionally, premature deaths associated with anthropogenic $PM_{2.5} + O_3$ were documented in China to be 1.2 million (Lelieveld et al., 2015) in 2010 (Figure 10). In comparison, our study has estimated 0.9 (95% CI: 0.7–1.1), 1.1 (95% CI: 0.9–1.3) and 0.6 (95% CI: 0.5–0.7) million deaths for CEDS, CAMS and ECLIPSE, respectively (Figure 10). This study estimates APDs due to anthropogenic $PM_{2.5} + O_3$ in India to be 0.6 (95% CI: 0.5–0.8), 0.7 (95% CI: 0.5–0.8), and 0.5 million (95% CI: 0.4–0.7) in 2015 for CEDS, CAMS, and ECLIPSE, respectively. These estimates are similar to those reported by Lelieveld et al. (2015), who estimated APDs associated with anthropogenic $PM_{2.5} + O_3$ in India to be 0.6 million in 2010 (Figure 10).

Previous study conducted by Lelieveld et al. (2015) estimated that, when exposure to anthropogenic $PM_{2.5} + O_3$, India and China accounted for 41% and 19% of total APD, respectively. Similarly, Silva et al. (2016)

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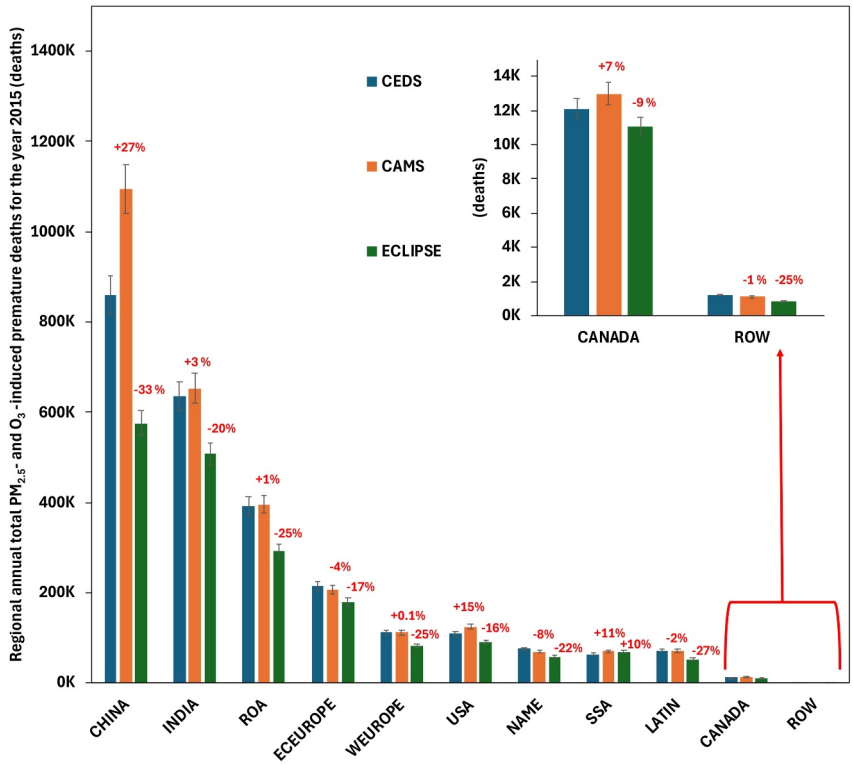


Figure 8. Combined regional annual total $PM_{2.5}$- and O_3-induced premature deaths for the year 2015 for Community Emissions Data System (CEDS) (blue), Copernicus Atmosphere Monitoring Service (CAMS) (orange) and Evaluating of the Climate and Air Quality Impacts of Short-Lived Pollutants (ECLIPSE) (green). Percentages in the bar plots are the percentage differences for CAMS and ECLIPSE using CEDS as a baseline. The error bars represent 95% confidence intervals.

found that in 2005, exposure to $PM_{2.5} + O_3$ for the year 2005 resulted in most of the mortality occurring in East Asia (45%) and India (20%).

The results of our study are consistent with previous studies when investigating both global natural and anthropogenic emissions. We found that the 6.8 million (95% CI: 6.0–7.6) APD reported by Xiong, Partha, et al. (2022) in 2015, using the CEDS emissions inventory were, while higher, comparable to our estimates of global premature deaths attributable to $PM_{2.5} + O_3$ exposure. Our estimates were 5.3 (95% CI: 4.6–6.4), 5.6 (95% CI: 4.9–6.7), and 4.7 (95% CI: 4.1–5.6) million for CEDS, CAMS and ECLIPSE respectively (Figure S8 in Supporting Information S1). However, previous studies reported significantly higher estimates. For instance, Burnett et al. (2018) reported 8.9 (95% CI: 7.5–10.3) million premature deaths in 2015 while Huang et al. (2020) had estimated 9.3 (95% CI: 8.3–10.2) million premature deaths in 2010. In 2015 Liu et al. (2021) estimated 1.8 million premature deaths in China. It was also found that emissions fell in China post-2013 due to strict policies (Figure 10), yet mortality rose 31% from 2005 to 2017 due to an aging population. Meanwhile, India, Africa, and Southeast Asia saw worsening pollution from urbanization, industrial growth, and biomass burning, while North America and Europe saw emission declines due to regulations. These trends explain why Lelieveld et al. (2015) estimated 3.3 million deaths in 2010. Liu et al. (2021) showed rising mortality despite lower emissions, and Burnett et al. (2018) estimated 8.9 million deaths in 2015, 30% higher than prior models.

The annual deaths of GBD 2019 for the year 2015 associated with NCD + LRI were 41.3 million, with our estimated ambient annual premature mortality attributable to $PM_{2.5}$ and O_3 exposure accounting for 13%, 14%, and 11%, for CESM, CAMS and ECLIPSE, respectively (GBD 2019 Risk Factors Collaborators, 2020; Xiong, Partha, et al., 2022). Similarly, the percentages of the annual YLL in 2015 from CESM, CAMS and ECLIPSE relative to the estimates of GBD 2019 in 2015 were 18%, 19% and 17%, respectively.

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Table 4
Comparisons of $PM_{2.5}$- and O_3-Induced Premature Death (APD) Rates (per 100,000 Population) Between Developed and Developing Countries in 2015

Cases	Species	Regions	2015 APD rates
CEDS	O_3	Developed	0.74 (0.46, 1.0)
		Developing	6.2 (4.0, 8.5)
	$PM_{2.5}$	Developed	1.6 (1.5, 1.8)
		Developing	16.9 (15.5, 19.9)
	$PM_{2.5} + O_3$	Developed	2.3 (1.9, 2.8)
		Developing	23.1 (19.5, 28.5)
CAMS	O_3	Developed	0.70 (0.43, 0.96)
		Developing	6.0 (3.9, 8.2)
	$PM_{2.5}$	Developed	1.6 (1.5, 1.8)
		Developing	19.6 (18.0, 23.0)
	$PM_{2.5} + O_3$	Developed	2.5 (2.1, 3.0)
		Developing	25.6 (23.0, 31.2)
ECLIPSE	O_3	Developed	0.54 (0.32, 0.72)
		Developing	3.7 (2.3, 5.3)
	$PM_{2.5}$	Developed	1.3 (1.2, 1.5)
		Developing	13.7 (12.5, 16.1)
	$PM_{2.5} + O_3$	Developed	1.9 (1.5, 2.2)
		Developing	17.4 (14.8, 21.2)

Note. Parenthesis shows the uncertainty and 95% confidence intervals. Developed countries include the USA, Canada, WEurope, and ROW; while developing countries refer to China, India, ROA, SSA, ECEurope, NAME and LATIN.

The disparities in estimates could stem from various factors, including differences in CTM, emissions inventories, and methodologies for assessing health burdens. Given that studies on human health burden assessments employ either the integrated exposure-response (IER) or GEMM, our utilization of GEMM may shed light on disparities from prior research as the GEMM method effectively addressed the limitations of the IER approach (Burnett et al., 2018). A limitation for GEMM and IER is the assumption of uniform toxicity across all $PM_{2.5}$ compositions. These models estimate mortality risk based on total mass concentration-without differentiating between chemical composition or source specific toxicity. This may obscure the regional differences in health impacts driven by variations in $PM_{2.5}$ composition (Burnett et al., 2018; Dominici et al., 2019; GBD 2019 Risk Factors Collaborators, 2020; Maji, 2020; Stanek et al., 2011). Assuming uniform toxicity may misrepresent health risks by ignoring differences in $PM_{2.5}$ compositions.

Maji (2020) found that premature mortality estimates using the GEMM models for China surpassed those that used IER models by 104% and 61%, respectively. Pope et al. (2018) highlighted that in heavily polluted regions, the IER might underestimate the health impacts linked to $PM_{2.5}$. This is relevant for regions such as India and China.

For example, Liu et al. (2021) utilized GEMM and estimated APD due to ambient $PM_{2.5}$ exposure for China was 1.4 million in 2005 and 1.8 million in 2017. In contrast, studies using IER reported 1.0 (Zhao et al., 2017) and 1.3 million APDs (Wang et al., 2018) for 2010. Comparing these to our estimates of 1.2 (95% CI: 1.1–1.4), 1.5 (95% CI: 1.4–1.7) and 1.1 (95% CI: 1.0–1.2) million APD attributable to ambient $PM_{2.5}$ for CEDS, CAMS and ECLIPSE, respectively, we observe that the IER models yield lower estimates.

A study by J. Liu et al. (2020), using Multi-resolution Emission Inventory for China (MEIC) and MIX emission inventory (Li et al., 2017) along with the GEMM and Weather Research and Forecasting model and the Community Multiscale Air Quality Modeling System, estimated APDs attributed to anthropogenic $PM_{2.5}$ as 1.6 (95% CI: 1.1–2.0) million in 2015. This estimate is significantly higher than the estimates in our study for China region: 0.6 million (95% CI: 0.5–0.7) from CEDS, 0.9 million (95% CI: 0.8–1.0) from CAMS, and 0.5 million (95% CI: 0.4–0.6) from ECLIPSE.

Regionally, the findings in our study indicate that India, China, and the ROA, experienced the highest proportion of premature deaths associated with $PM_{2.5} + O_3$ exposure, at 33.8% (95% CI: 33.7%–33.9%), 38.9% (95% CI: 33.7%–47.1%) and 30.2% (95% CI: 30.0%–30.4%) of premature deaths for CEDS, CAMS and ECLIPSE respectively. We found that majority of $PM_{2.5}$- and O_3-induced premature deaths occur in Asia and Africa. However, due to limited ground observational data in these regions, there may be over or underestimations in air quality impacts and mortality calculations (Salameh et al., 2016; Shami et al., 2022).

Future human health impact studies in the SSA and NAME regions are crucial, given our estimates of total global $PM_{2.5} + O_3$ premature deaths at 5.5% (95% CI: 5.1–5.7), 5.0% (95% CI: 4.2–5.8), 6.0 (95% CI: 5.3–7.8) for CEDS, CAMS and ECLIPSE. In 2015, the NAME super-region exhibited the highest death rates due to air pollution (Abbasi-Kangevari et al., 2023). Similarly in 2019 SSA region estimated 16.3% deaths due to air pollution (Fisher et al., 2021a, 2021b).

Huang et al. (2021) had highlighted that large share of residential energy in low and middle-income countries, all categorized as "Developing" in this study, relies on solid biofuel for cooking, heating, and lighting. WHO estimates that 90% deaths due to outdoor air pollution occur mainly in Asia and Africa. It is estimated that 55.3% of the world's population are exposed to increased levels of $PM_{2.5}$ (Shaddick et al., 2020) and this is expected to worsen due to globalization.

For example, India's major cities, located within the Indo-Gangetic Plain region, experience multiday air pollution events annually due to high population and pollution levels, significantly contributing to $PM_{2.5}$-related premature

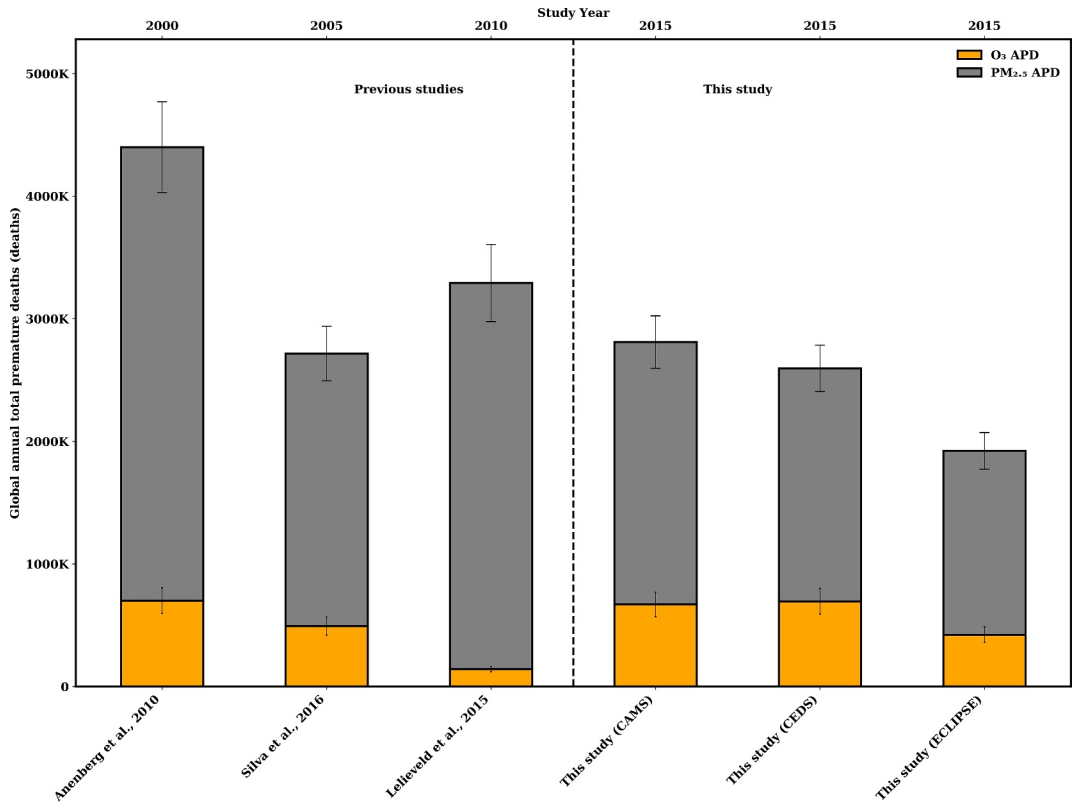


Figure 9. Comparisons of global total annual total premature deaths (APDs) associated with anthropogenic sources, APD associated with O_3 (orange), and APD associated with $PM_{2.5}$ (gray). Total APD is including both $PM_{2.5}$ + O_3. The error bars show 95% confidence intervals of estimated premature deaths resulted from $PM_{2.5}$ + O_3 exposures.

deaths (Asher Ghertner, 2021; David et al., 2019; Roozitalab et al., 2021). Similarly, in China, the regions most impacted by $PM_{2.5}$ pollution include the North China Plain and the Yangtze River Delta (Zhai et al., 2019; Zhang et al., 2022), where rapid urbanization and industrial activities lead to severe air quality issues, resulting in high rates of premature deaths related to $PM_{2.5}$ exposure.

This burden disproportionately affects the most vulnerable populations, including low-income individuals and those in developing nations (Galimova et al., 2022; Gardner-Frolick et al., 2022). Our study shows that 79.5% (CEDS), 81.2% (CAMS), and 77.6% (ECLIPSE) of premature deaths due to anthropogenic activities are associated with Asia and Africa continent. Figure 11 shows an illustration of percentage for Asia and Africa in CEDS, CAMS and ECLIPSE. In the context of Asian and African nations (China, India, ROA, NAME and SSA) exclusively, this study reveals a consistent pattern across the three inventories.

Lastly, we acknowledge the discrepancies with previous studies due to different methodologies and emission inventories, as previously mentioned. Despite this, our study shows a notable trend: estimated global $PM_{2.5}$ mortality is three times higher than the O_3 mortality, indicating that $PM_{2.5}$ is the primary contributor of the outdoor air pollution-related global health burden. This aligns with a previous study by Anenberg et al. (2010), which reported that their estimated $PM_{2.5}$ mortality was five times higher than the O_3 mortality.

We also recognize that, in addition to the differences in emission inventories explored here, surface $PM_{2.5}$ + O_3 and premature deaths attributed to $PM_{2.5}$ + O_3, are model-dependent and influenced by the GBD study utilized. If available, it is recommended to use localized emission inventories, recognizing that no inventory is entirely precise (Madrazo et al., 2018), yet such an approach could enhance emission analysis. The limitations in data and methodologies utilized for developing these inventories, as highlighted by Miller et al. (2006) can significantly impact air quality management.

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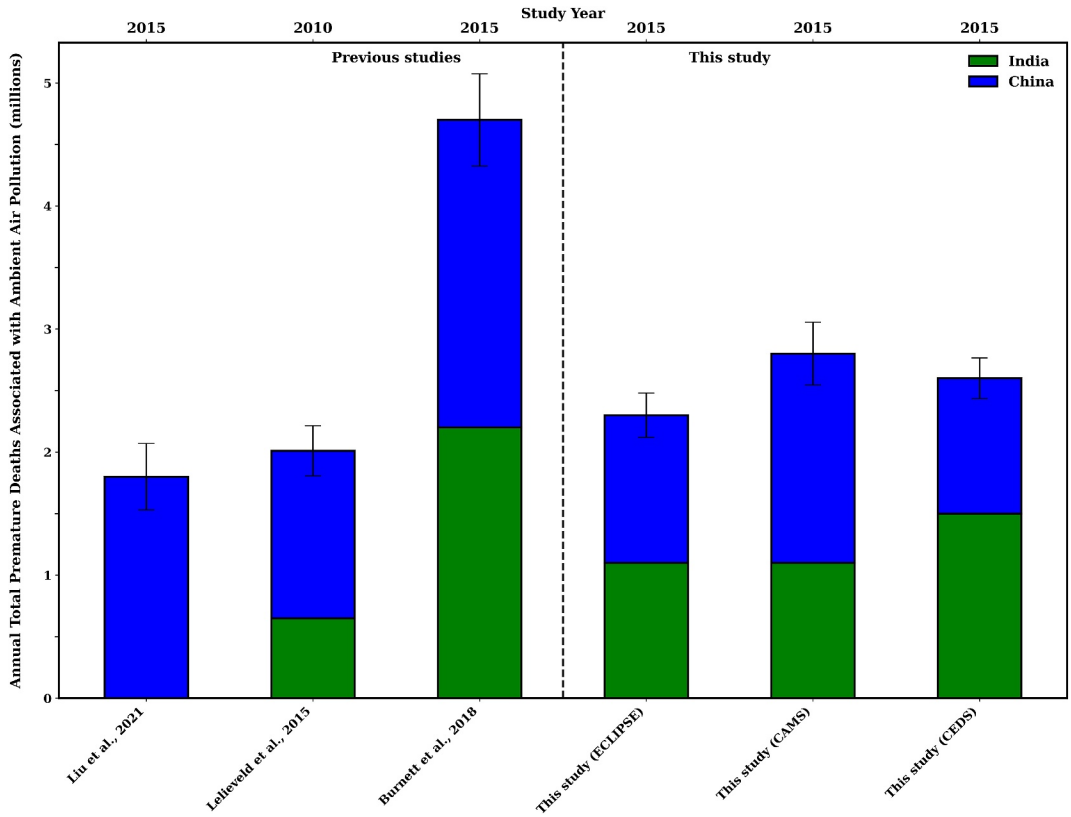


Figure 10. Comparisons of global annual total premature deaths associated with ambient $PM_{2.5}$ for India and China. India and China are represented by green and blue bars respectively. The *y* axis is in millions. The error bars show 95% confidence intervals of estimated premature deaths.

4. Conclusions

We investigated how differences in emissions inventories (CEDS, CAMS, and ECLIPSE) affect air quality and human health impact estimates in 2015 using CAM-chem. This study is the first study contrasting these three emission inventories and quantifying the global human health impacts attributable to $PM_{2.5}$ and O_3 exposure due to anthropogenic activities.

Due to existing uncertainties in anthropogenic emission inventories, this significantly impacts the accuracy of health impact assessments. As evidenced by our intercomparisons using CEDS as the baseline, we found

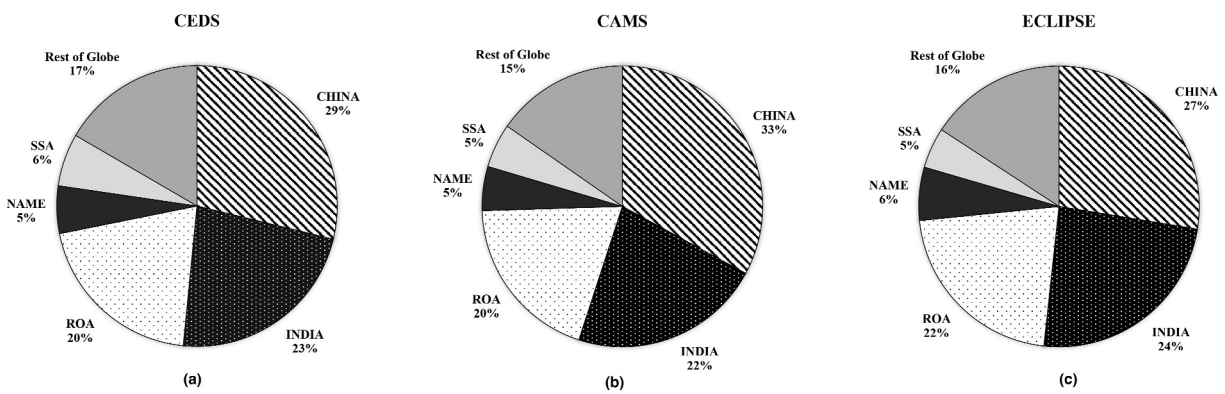


Figure 11. Pie chart of percentages of $PM_{2.5} + O_3$ annual total premature deaths for Rest of Globe (LATIN, ECEUROPE, WEUROPE, USA, CANADA and Rest of the world), China, NAME, Rest of Asia, India and SSA for (a) Community Emissions Data System, (b) Copernicus Atmosphere Monitoring Service and (c) Evaluating of the Climate and Air Quality Impacts of Short-Lived Pollutants.

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substantial discrepancies in health impact estimates when using different emission inventories. For example, we found that ECLIPSE APDs and air quality impacts were significantly lower. We had found that $PM_{2.5}$- and O_3-induced premature deaths for the year 2015 to be 2.5 (95% CI: 2.1–3.1), 2.8 (95% CI: 2.4–3.4), and 1.9 (95% CI: 1.6–2.4) million for CEDS, CAMS and ECLIPSE respectively. The regions with the highest APDs are China, India, and ROA. The corresponding YLL associated with $PM_{2.5} + O_3$ in 2015 were calculated to be 55.4 (95% CI: 50.6–64.1), 62.7 (95% CI: 54.5–70.6) and 62.1 (95% CI: 45.2–85.4) million years for CEDS, CAMS, and ECLIPSE, respectively. This illustrates how differences in emission data can lead to uncertainties in estimating health impacts. Upon validating the model, we found a much larger negative NMB for Europe, India, and China using the ECLIPSE emissions inventory, indicating a significant underprediction by the model. We found that discrepancies in emissions inventories contributed to differences in human health impact estimates, highlighting the critical need for accurate emission inventories to improve health impact assessments and inform effective air quality policies. This issue is pronounced in China and India, two regions with large populations, where existing underestimations contribute significantly to bias in our estimates of the associated health burden.

Our findings highlight the differences when using commonly used emission inventories to investigate air quality and human health impacts. Additionally, we identified significant disparities in the impact of air pollution on premature deaths due to $PM_{2.5}$ and O_3 exposure. Developing nations experiencing an average impact that it is on average 11 times higher than that in Developed nations using CEDS, CAMS and ECLIPSE inventories. Our study not only offers valuable scientific evidence for global policymakers in their efforts to mitigate the impact of air quality on human health, but also highlights the importance of improving emissions inventories to ensure more accurate health impact assessments.

Given the existing discrepancies across inventories, an multi-model ensemble approach may help reduce uncertainties by using multiple models with different assumptions, and chemical mechanisms (Fiore et al., 2009; Pachón et al., 2024; Thornhill et al., 2021; Van Donkelaar et al., 2016). Using a multi-model ensemble may help mitigate individual model biases. By averaging across simulations, it has the potential to align more closely with ground-based observations and GBD 2019 mortality estimates, thus offering a more robust assessment of population exposure and the associated health burden than relying on a single emission inventory (Deroubaix et al., 2024; Pozzer et al., 2023). Improving emissions estimates in regions with limited observational data remains a key priority and future efforts would benefit from integrating ground-based observations with satellite observations (Miyazaki et al., 2020). Furthermore, it is important to evaluate how different inventories perform across regions, as their accuracy depends on data availability and emission sources.

Lastly, interdisciplinary collaboration between atmospheric scientists, public health experts, and policymakers is crucial for translating air quality modeling results into actionable public health interventions (Abera et al., 2024; Fisher et al., 2021a, 2021b; Holloway et al., 2024). Future studies should also consider exploring approaches that account for regional health vulnerabilities, particularly in the Global South, where pollution burdens are disproportionately high. Addressing these gaps is critical to implementing effective air quality regulations and reducing premature deaths globally.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

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Data Availability Statement

Data of CESM CAM6-Chem model-simulated $PM_{2.5}$ and O_3 concentrations and associated premature deaths in 2015 are available publicly at Salah (2025).

References

Abbasi-Kangevari, M., Malekpour, M. R., Masinaei, M., Moghaddam, S. S., Ghamari, S. H., Abbasi-Kangevari, Z., et al. (2023). Effect of air pollution on disease burden, mortality, and life expectancy in North Africa and the Middle East: A systematic analysis for the global burden of disease study 2019. *The Lancet Planetary Health*, *7*(5), e358–e369. https://doi.org/10.1016/S2542-5196(23)00053-0

Abera, A., Friberg, J., Isaxon, C., Jerrett, M., Malmqvist, E., Sjöström, C., et al. (2024). Air quality in Africa: Public health implications. *Annual Review of Public Health*, *42*(1), 34–210. https://doi.org/10.1146/annurev-publhealth-100119-113802

Amann, M., Bertok, I., Borken-Kleefeld, J., Cofala, J., Heyes, C., Höglund-Isaksson, L., et al. (2011). Cost-effective control of air quality and greenhouse gases in Europe: Modeling and policy applications. *Environmental Modelling & Software*, *26*(12), 1489–1501. https://doi.org/10.1016/j.envsoft.2011.07.012

Andrew, R. M. (2020). A comparison of estimates of global carbon dioxide emissions from fossil carbon sources. *Earth System Science Data*, *12*(2), 1437–1465. https://doi.org/10.5194/essd-12-1437-2020

Anenberg, S. C., Horowitz, L. W., Tong, D. Q., & West, J. J. (2010). An estimate of the global burden of anthropogenic ozone and fine particulate matter on premature human mortality using atmospheric modeling. *Environmental Health Perspectives*, *118*(9), 1189–1195. https://doi.org/10.1289/ehp.0901220

Ansari, T. U., Ojha, N., Chandrasekar, R., Balaji, C., Singh, N., & Gunthe, S. S. (2016). Competing impact of anthropogenic emissions and meteorology on the distribution of trace gases over Indian region. *Journal of Atmospheric Chemistry*, *73*(4), 363–380. https://doi.org/10.1007/s10874-016-9331-y

Archer-Nicholls, S., Carter, E., Kumar, R., Xiao, Q., Liu, Y., Frostad, J., et al. (2016). The regional impacts of cooking and heating emissions on ambient air quality and disease burden in China. *Environmental Science and Technology*, *50*(17), 9416–9423. https://doi.org/10.1021/acs.est.6b02533

Asher Ghertner, D. (2021). Postcolonial atmospheres: Air's coloniality and the climate of enclosure. *Annals of the American Association of Geographers*, *111*(5), 1483–1502. https://doi.org/10.1080/24694452.2020.1823201

Beverland, I. J., Cohen, G. R., Heal, M. R., Carder, M., Yap, C., Robertson, C., et al. (2012). A comparison of short-term and long-term air pollution exposure associations with mortality in two cohorts in Scotland. *Environmental Health Perspectives*, *120*(9), 1280–1285. https://doi.org/10.1289/ehp.1104509

Bouarar, I., Gaubert, B., Brasseur, G. P., Steinbrecht, W., Doumbia, T., Tilmes, S., et al. (2021). Ozone anomalies in the free troposphere during the COVID-19 pandemic. *Geophysical Research Letters*, *48*(16), 1–11. https://doi.org/10.1029/2021GL094204

Bran, S. H., Macatangay, R., Surapipith, V., Chotamonsak, C., Chantara, S., Han, Z., & Li, J. (2022). Surface PM2.5 mass concentrations during the dry season over northern Thailand: Sensitivity to model aerosol chemical schemes and the effects on regional meteorology. *Atmospheric Research*, *277*, 106303. https://doi.org/10.1016/j.atmosres.2022.106303

Bui, L. T., Nguyen, N. H. T., & Nguyen, P. H. (2023). Chronic and acute health effects of PM2.5 exposure and the basis of pollution control targets. *Environmental Science and Pollution Research*, *30*(33), 79937–79959. https://doi.org/10.1007/s11356-023-27936-9

Burnett, R., Chen, H., Szyszkowicz, M., Fann, N., Hubbell, B., Pope, C. A., et al. (2018). Global estimates of mortality associated with longterm exposure to outdoor fine particulate matter. *Proceedings of the National Academy of Sciences of the United States of America*, *115*(38), 9592–9597. https://doi.org/10.1073/pnas.1803222115

Center for International Earth Science Information Network – CIESIN – Columbia University. (2018). *Gridded population of the world, version 4 (GPWv4): Population density, revision 11*. NASA Socioeconomic Data and Applications Center (SEDAC).

Cheyenne Supercomputer. (2021). HPE/SGI ICE XA system. Computational and information systems laboratory. *National Center for Atmospheric Research*. https://doi.org/10.5065/D6RX99HX

Choulga, M., Janssens-maenhout, G., Super, I., Agusti-panareda, A., Bousserez, N., Crippa, M., et al. (2020). Earth system modelling and data assimilation. *Earth System Science Data*, 1–37. https://doi.org/10.5281/zenodo.3712339.35

Chowdhury, S., Dey, S., Guttikunda, S., Pillarisetti, A., Smith, K. R., & Girolamo, L. D. (2019). Indian annual ambient air quality standard is achievable by completely mitigating emissions from household sources. *Proceedings of the National Academy of Sciences of the United States of America*, *166*(22), 10711–10716. https://doi.org/10.1073/pnas.1900888116

Contini, D., Vecchi, R., & Viana, M. (2018). Carbonaceous aerosols in the atmosphere. *Atmosphere*, *9*(5), 1–8. https://doi.org/10.3390/atmos9050181

Crippa, M., Guizzardi, D., Muntean, M., Schaaf, E., Dentener, F., van Aardenne, J. A., et al. (2018). Gridded emissions of air pollutants for the period 1970–2012 within EDGAR v4.3.2. *Earth System Science Data*, *10*(4), 1987–2013. https://doi.org/10.5194/essd-10-1987-2018

Crippa, M., Janssens-Maenhout, G., Guizzardi, D., Van Dingenen, R., & Dentener, F. (2019). Contribution and uncertainty of sectorial and regional emissions to regional and global PM2.5 health impacts. *Atmospheric Chemistry and Physics*, *19*(7), 5165–5186. https://doi.org/10.5194/acp-19-5165-2019

Danabasoglu, G., Lamarque, J. F., Bacmeister, J., Bailey, D. A., DuVivier, A. K., Edwards, J., et al. (2020). The community Earth system model version 2 (CESM2). *Journal of Advances in Modeling Earth Systems*, *12*(2), 1–35. https://doi.org/10.1029/2019MS001916

David, L. M., Ravishankara, A. R., Kodros, J. K., Pierce, J. R., Venkataraman, C., & Sadavarte, P. (2019). Premature mortality due to PM2.5 over India: Effect of atmospheric transport and anthropogenic emissions. *GeoHealth*, *3*(1), 2–10. https://doi.org/10.1029/2018GH000169

Deroubaix, A., Hoelzemann, J. J., Ynoue, R. Y., Toledo de Almeida Albuquerque, T., Alves, R. C., de Fatima Andrade, M., et al. (2024). Intercomparison of air quality models in a megacity: Toward an operational ensemble forecasting system for São Paulo. *Journal of Geophysical Research: Atmospheres*, *129*(1), e2022JD038179. https://doi.org/10.1029/2022JD038179

Dominici, F., Schwartz, J., Di, Q., Braun, D., Choirat, C., & Zanobetti, A. (2019). Assessing adverse health effects of long-term exposure to low levels of ambient air pollution: Phase 1. Retrieved from www.healtheffects.org

Dong, G. H., Zhang, P., Sun, B., Zhang, L., Chen, X., Ma, N., et al. (2012). Long-term exposure to ambient air pollution and respiratory disease mortality in Shenyang, China: A 12-year population-based retrospective cohort study. *Respiration*, *84*(5), 360–368. https://doi.org/10.1159/000332930

Elguindi, N., Granier, C., Stavrakou, T., Darras, S., Bauwens, M., Cao, H., et al. (2020). Intercomparison of magnitudes and trends in anthropogenic surface emissions from Bottom-Up inventories, top-down estimates, and emission scenarios. *Earth's Future*, *8*(8), 1–20. https://doi.org/10.1029/2020EF001520

Emmons, L. K., Schwantes, R. H., Orlando, J. J., Tyndall, G., Kinnison, D., Lamarque, J. F., et al. (2020). The chemistry mechanism in the community Earth system model version 2 (CESM2). *Journal of Advances in Modeling Earth Systems*, *12*(4), 1–21. https://doi.org/10.1029/2019MS001882

EPA. (2020). ENVIRONMENTAL PROTECTION AGENCY 40 CFR part 50 review of the national ambient air quality standards for particulate matter. Retrieved from https://cfpub.epa.gov/ncea/

Fiore, A. M., Dentener, F. J., Wild, O., Cuvelier, C., Schultz, M. G., Hess, P., et al. (2009). Multimodel estimates of intercontinental source-receptor relationships for ozone pollution. *Journal of Geophysical Research*, *114*(4). https://doi.org/10.1029/2008JD010816

Fisher, S., Bellinger, D. C., Cropper, M. L., Kumar, P., Binagwaho, A., Koudenoukpo, J. B., et al. (2021a). Air pollution and development in Africa: Impacts on health, the economy, and human capital. *The Lancet Planetary Health*, *5*(10), e681–e688. https://doi.org/10.1016/S2542-5196(21)00201-1

Fisher, S., Bellinger, D. C., Cropper, M. L., Kumar, P., Binagwaho, A., Koudenoukpo, J. B., et al. (2021b). Global public health and the air pollution and development in Africa: Impacts on health, the economy, and human capital. *Articles Lancet Planet Health*, *5*(10), e681–e688. https://doi.org/10.1016/s2542-5196(21)00201-1

Galimova, T., Ram, M., & Breyer, C. (2022). Mitigation of air pollution and corresponding impacts during a global energy transition towards 100% renewable energy system by 2050. *Energy Reports*, *8*, 14124–14143. https://doi.org/10.1016/j.egyr.2022.10.343

24711403, 2025, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GH001223 by Halima Salah , Wiley Online Library on [18/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Gardner-Frolick, R., Boyd, D., & Giang, A. (2022). Selecting data analytic and modeling methods to support air pollution and environmental justice investigations: A critical review and guidance framework. *Environmental Science and Technology*, *56*(5), 2843–2860. https://doi.org/10.1021/acs.est.1c01739

Gaubert, B., Bouarar, I., Doumbia, T., Liu, Y., Stavrakou, T., Deroubaix, A., et al. (2021). Global changes in secondary atmospheric pollutants during the 2020 COVID-19 pandemic. *Journal of Geophysical Research: Atmospheres*, *126*(8), 1–22. https://doi.org/10.1029/2020JD034213

GBD 2019 Risk Factors Collaborators, Aravkin, A. Y., Zheng, P., Abbafati, C., Abbas, K. M., Abbasi-Kangevari, M., et al. (2020). Global burden of 87 risk factors in 204 countries and territories, 1990–2019: A systematic analysis for the global burden of disease study 2019. *The Lancet*, *396*(10258), 1223–1249. https://doi.org/10.1016/S0140-6736(20)30752-2

Ge, Y., Vieno, M., Stevenson, D. S., Wind, P., & Heal, M. R. (2023). Global sensitivities of reactive N and S gas and particle concentrations and deposition to precursor emissions reductions. *Atmospheric Chemistry and Physics*, *23*(11), 6083–6112. https://doi.org/10.5194/acp-23-6083-2023

Gettelman, A., Mills, M. J., Kinnison, D. E., Garcia, R. R., Smith, A. K., Marsh, D. R., et al. (2019). The whole atmosphere community climate model version 6 (WACCM6). *Journal of Geophysical Research: Atmospheres*, *124*(23), 12380–12403. https://doi.org/10.1029/2019JD030943

Granier, C., Darras, S., Denier van der Gon, H., Doubalova, J., Elguindi, N., Galle, B., et al. (2019). The copernicus atmosphere monitoring service global and regional emissions. https://doi.org/10.24380/d0bn-kx16

Guenther, A. B., Jiang, X., Heald, C. L., Sakulyanontvittaya, T., Duhl, T., Emmons, L. K., & Wang, X. (2012). The model of emissions of gases and aerosols from nature version 2.1 (MEGAN2.1): An extended and updated framework for modeling biogenic emissions. *Geoscientific Model Development*, *5*(6), 1471–1492. https://doi.org/10.5194/gmd-5-1471-2012

Hoesly, R. M., Smith, S. J., Feng, L., Klimont, Z., Janssens-Maenhout, G., Pitkanen, T., et al. (2018). Historical (1750-2014) anthropogenic emissions of reactive gases and aerosols from the community emissions data system (CEDS). *Geoscientific Model Development*, *11*(1), 369–408. https://doi.org/10.5194/gmd-11-369-2018

Holloway, T., Miller, D., Anenberg, S., Diao, M., Duncan, B., Fiore, A. M., et al. (2024). Satellite monitoring for air quality and health. *Annual Review of Biomedical Data Science*, *4*(1), 417–447. https://doi.org/10.1146/annurev-biodatasci-110920-093120

Horowitz, L. W. (2006). Past, present and future concentrations of tropospheric ozone and aerosols: Methodology, ozone evaluation, and sensitivity to aerosol wet removal. *Journal of Geophysical Research*, *111*(22). https://doi.org/10.1029/2005JD006937

Huang, Y., Partha, D. B., Harper, K., & Heyes, C. (2021). Impacts of global solid biofuel stove emissions on ambient air quality and human health. *GeoHealth*, *5*(3), 1–16. https://doi.org/10.1029/2020GH000362

Huang, Y., Unger, N., Harper, K., & Heyes, C. (2020). Global climate and human health effects of the gasoline and diesel vehicle fleets. *GeoHealth*, *4*(3), 1–13. https://doi.org/10.1029/2019GH000240

Huang, Y., Unger, N., Storelvmo, T., Harper, K., Zheng, Y., & Heyes, C. (2018). Global radiative effects of solid fuel cookstove aerosol emissions. *Atmospheric Chemistry and Physics*, *18*(8), 5219–5233. https://doi.org/10.5194/acp-18-5219-2018

Huang, Y., Wu, S., Kramer, L. J., Helmig, D., & Honrath, R. E. (2017). Surface ozone and its precursors at summit, Greenland: Comparison between observations and model simulations. *Atmospheric Chemistry and Physics*, *17*(23), 14661–14774. https://doi.org/10.5194/acp-17-14661-2017

Iacono, M. J., Delamere, J. S., Mlawer, E. J., Shephard, M. W., Clough, S. A., & Collins, W. D. (2008). Radiative forcing by long-lived greenhouse gases: Calculations with the AER radiative transfer models. *Journal of Geophysical Research*, *113*(13), 2–9. https://doi.org/10.1029/2008JD009944

IPCC. (2023). Summary for policymakers: Synthesis report. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, 1–34.

Jo, D. S., Hodzic, A., Emmons, L. K., Tilmes, S., Schwantes, R. H., Mills, M. J., et al. (2021). Future changes in isoprene-epoxydiol-derived secondary organic aerosol (IEPOX SOA) under the shared socioeconomic pathways: The importance of physicochemical dependency. *Atmospheric Chemistry and Physics*, *21*(5), 3395–3425. https://doi.org/10.5194/acp-21-3395-2021

Jo, D. S., Tilmes, S., Emmons, L. K., Wang, S., & Vitt, F. (2023). A new simplified parameterization of secondary organic aerosol in the community Earth system model version 2 (CESM2; CAM6.3). *Geoscientific Model Development*, *16*(13), 3893–3906. https://doi.org/10.5194/gmd-16-3893-2023

Klimont, Z., Heyes, C., Hoglund-Isaksson, L., Lindl, F., Kim, Y., Rafaj, P., et al. (2025). Global gridded anthropogenic emissions of air pollutants and methane for the period 1990-2050. *Zenodo*. https://doi.org/10.5281/zenodo.14748815

Klimont, Z., Kupiainen, K., Heyes, C., Purohit, P., Cofala, J., Rafaj, P., et al. (2017). Global anthropogenic emissions of particulate matter including black carbon. *Atmospheric Chemistry and Physics*, *17*(14), 8681–8723. https://doi.org/10.5194/acp-17-8681-2017

Kuenen, J., Dellaert, S., Visschedijk, A., Jalkanen, J. P., Super, I., & Denier Van Der Gon, H. (2022). CAMS-REG-v4: A state-of-the-art high-resolution European emission inventory for air quality modelling. *Earth System Science Data*, *14*(2), 491–515. https://doi.org/10.5194/essd-14-491-2022

Kurokawa, J., & Ohara, T. (2020). Long-term historical trends in air pollutant emissions in Asia: Regional emission inventory in ASia (REAS) version 3. *Atmospheric Chemistry and Physics*, *20*(21), 12761–12793. https://doi.org/10.5194/acp-20-12761-2020

Lamarque, J. F., Emmons, L. K., Hess, P. G., Kinnison, D. E., Tilmes, S., Vitt, F., et al. (2012). CAM-chem: Description and evaluation of interactive atmospheric chemistry in the community Earth system model. *Geoscientific Model Development*, *5*(2), 369–411. https://doi.org/10.5194/gmd-5-369-2012

Lawrence, D., Fisher, R., Koven, C., Oleson, K., Swenson, S., Bonan, G., et al. (2019). The community land model version 5: Description of new features, benchmarking, and impact of forcing uncertainty. *Journal of Advances in Modeling Earth Systems*, *11*(12), 4245–4287. https://doi.org/10.1029/2018MS001583

Lelieveld, J., Evans, J. S., Fnais, M., Giannadaki, D., & Pozzer, A. (2015). The contribution of outdoor air pollution sources to premature mortality on a global scale. *Nature*, *525*(7569), 367–371. https://doi.org/10.1038/nature15371

Leung, D. M., Kok, J. F., Li, L., Lawrence, D. M., Mahowald, N. M., Tilmes, S., & Kluzek, E. (2025). A global dust emission dataset for estimating dust radiative forcings in climate models. *Atmospheric Chemistry and Physics*, *25*(4), 2311–2331. https://doi.org/10.5194/acp-25-2311-2025

Leung, D. M., Kok, J. F., Li, L., Mahowald, N. M., Lawrence, D. M., Tilmes, S., et al. (2024). A new process-based and scale-aware desert dust emission scheme for global climate models – Part II: Evaluation in the community Earth system model version 2 (CESM2). *Atmospheric Chemistry and Physics*, *24*(4), 2287–2318. https://doi.org/10.5194/acp-24-2287-2024

Li, M., Zhang, Q., Kurokawa, J. I., Woo, J. H., He, K., Lu, Z., et al. (2017). MIX: A mosaic Asian anthropogenic emission inventory under the international collaboration framework of the MICS-Asia and HTAP. *Atmospheric Chemistry and Physics*, *17*(2), 935–963. https://doi.org/10.5194/acp-17-935-2017

24711403, 2025, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GH001223 by Halima Salah , Wiley Online Library on [18/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Liu, J., Mauzerall, D. L., Chen, Q., Zhang, Q., Song, Y., Peng, W., et al. (2016a). Air pollutant emissions from Chinese households: A major and underappreciated ambient pollution source. *Proceedings of the National Academy of Sciences of the United States of America*, *113*(28), 7756–7761. https://doi.org/10.1073/pnas.1604537113

Liu, J., Zheng, Y., Geng, G., Hong, C., Li, M., Li, X., et al. (2020a). Decadal changes in anthropogenic source contribution of PM2.5 pollution and related health impacts in China, 1990-2015. *Atmospheric Chemistry and Physics*, *20*(13), 7783–7799. https://doi.org/10.5194/acp-20-7783-2020

Liu, M., Saari, R. K., Zhou, G., Li, J., Han, L., & Liu, X. (2021). Recent trends in premature mortality and health disparities attributable to ambient PM2.5 exposure in China: 2005–2017. *Environmental Pollution*, *279*, 116882. https://doi.org/10.1016/j.envpol.2021.116882

Liu, X., Ma, P. L., Wang, H., Tilmes, S., Singh, B., Easter, R. C., et al. (2016b). Description and evaluation of a new four-mode version of the modal aerosol module (MAM4) within version 5.3 of the community atmosphere model. *Geoscientific Model Development*, *9*(2), 505–522. https://doi.org/10.5194/gmd-9-505-2016

Liu, Y., Dong, X., Wang, M., Emmons, L., Liu, Y., Liang, Y., et al. (2020b). Analysis of secondary organic aerosol simulation bias in the community earth system model (CESM2.1). *Atmospheric Chemistry and Physics*, 1–24. https://doi.org/10.5194/acp-2020-1182

Lund, M. T., Aamaas, B., Stjern, C. W., Klimont, Z., Berntsen, T. K., & Samset, B. H. (2020). A continued role of short-lived climate forcers under the shared socioeconomic pathways. *Earth System Dynamics*, *11*(4), 977–993. https://doi.org/10.5194/esd-11-977-2020

Lund, M. T., Myhre, G., Skeie, R. B., Samset, B. H., & Klimont, Z. (2023). Implications of differences between recent anthropogenic aerosol emission inventories for diagnosed AOD and radiative forcing from 1990 to 2019. *Atmospheric Chemistry and Physics*, *23*(12), 6647–6662. https://doi.org/10.5194/acp-23-6647-2023

Madrazo, J., Clappier, A., Belalcazar, L. C., Cuesta, O., Contreras, H., & Golay, F. (2018). Screening differences between a local inventory and the emissions database for global atmospheric research (EDGAR). *Science of the Total Environment*, *631–632*, 934–941. https://doi.org/10.1016/j.scitotenv.2018.03.094

Maji, K. J. (2020). Substantial changes in PM2.5 pollution and corresponding premature deaths across China during 2015–2019: A model prospective. *Science of the Total Environment*, *729*, 138838. https://doi.org/10.1016/j.scitotenv.2020.138838

McDuffie, E. E., Smith, S. J., O'Rourke, P., Tibrewal, K., Venkataraman, C., Marais, E. A., et al. (2020). A global anthropogenic emission inventory of atmospheric pollutants from sector- and fuel-specific sources (1970-2017): An application of the community emissions data system (CEDS). *Earth System Science Data*, *12*(4), 3413–3442. https://doi.org/10.5194/essd-12-3413-2020

Miller, C. A., Hidy, G., Hales, J., Kolb, C. E., Werner, A. S., Haneke, B., et al. (2006). Air emission inventories in North America: A critical assessment. *Journal of the Air and Waste Management Association*, *56*(8), 1115–1129. https://doi.org/10.1080/10473289.2006.10464540

Miyazaki, K., Bowman, K., Sekiya, T., Eskes, H., Boersma, F., Worden, H., et al. (2020). Updated tropospheric chemistry reanalysis and emission estimates, TCR-2, for 2005-2018. *Earth System Science Data*, *12*(3), 2223–2259. https://doi.org/10.5194/essd-12-2223-2020

Nawaz, M. O., Henze, D. K., Anenberg, S. C., Braun, C., Miller, J., & Pronk, E. (2023). A source apportionment and emission scenario assessment of PM2.5- and O3-Related health impacts in G20 countries. *GeoHealth*, *7*(1), 1–23. https://doi.org/10.1029/2022GH000713

O'Rourke, P. R., Smith, S. J., Mott, A., Ahsan, H., McDuffie, E. E., Crippa, M., et al. (2021). CEDS v_2021_04_21 release emission data. *Zenodo*.

Pachón, J. E., Opazo, M. A., Lichtig, P., Huneeus, N., Bouarar, I., Brasseur, G., et al. (2024). Air quality modeling intercomparison and multiscale ensemble chain for Latin America. *Geoscientific Model Development*, *17*(20), 7467–7512. https://doi.org/10.5194/gmd-17-7467-2024

Pino-Cortés, E. (2021). Processing methodology of global anthropogenic emissions for air quality modeling. *MethodsX*, *8*, 101505. https://doi.org/10.1016/j.mex.2021.101505

Placet, M., Mann, C. O., Gilbert, R. O., & Niefer, M. J. (2000). Emissions of ozone precursors from stationary sources: A critical review. *Atmospheric Environment*, *34*(12–14), 2183–2204. https://doi.org/10.1016/S1352-2310(99)00464-1

Pope, C. A. (2000). Epidemiology of fine particulate air pollution and human health: Biologic mechanisms and who's at risk? *Environmental Health Perspectives*, *108*(4), 713–723. https://doi.org/10.2307/3454408

Pope, C. A., Cohen, A. J., & Burnett, R. T. (2018). Cardiovascular disease and fine particulate matter lessons and limitations of an integrated exposure-response approach. *Circulation Research*, *122*(12), 1645–1647. https://doi.org/10.1161/CIRCRESAHA.118.312956

Pozzer, A., Anenberg, S. C., Dey, S., Haines, A., Lelieveld, J., & Chowdhury, S. (2023). Mortality attributable to ambient air pollution: A review of global estimates. *GeoHealth*, *7*(1), e2022GH000711. https://doi.org/10.1029/2022GH000711

Roberts, G., & Wooster, M. J. (2021). Global impact of landscape fire emissions on surface level PM2.5 concentrations, air quality exposure and population mortality. *Atmospheric Environment*, *252*(July 2020), 118210. https://doi.org/10.1016/j.atmosenv.2021.118210

Roozitalab, B., Carmichael, G. R., & Guttikunda, S. K. (2021). Improving regional air quality predictions in the indo-gangetic plain - Case study of an intensive pollution episode in November 2017. *Atmospheric Chemistry and Physics*, *21*(4), 2837–2860. https://doi.org/10.5194/acp-21-2837-2021

Salah, H. (2025). Impact of anthropogenic emission estimates on air quality and human health effects [Dataset]. *Figshare*. https://doi.org/10.6084/m9.figshare.28745315.v2

Salameh, T., Sauvage, S., Afif, C., Borbon, A., & Locoge, N. (2016). Source apportionment vs. emission inventories of non-methane hydrocarbons (NMHC) in an urban area of the Middle East: Local and global perspectives. *Atmospheric Chemistry and Physics*, *16*(5), 3595–3607. https://doi.org/10.5194/acp-16-3595-2016

Sarkodie, S. A., Strezov, V., Jiang, Y., & Evans, T. (2019). Proximate determinants of particulate matter (PM2.5) emission, mortality and life expectancy in Europe, Central Asia, Australia, Canada and the US. *Science of the Total Environment*, *683*, 489–497. https://doi.org/10.1016/j.scitotenv.2019.05.278

Schultz, M. G., Schröder, S., Lyapina, O., Cooper, O. R., Galbally, I., Petropavlovskikh, I., et al. (2017). Tropospheric ozone assessment report: Database and metrics data of global surface ozone observations. *Elementa*, *5*, 58. https://doi.org/10.1525/elementa.244

Shaddick, G., Thomas, M. L., Mudu, P., Ruggeri, G., & Gumy, S. (2020). Half the world's population are exposed to increasing air pollution. *Npj Climate and Atmospheric Science*, *3*(1), 1–6. https://doi.org/10.1038/s41612-020-0124-2

Shami, A. A., Aawar, E. A., Baayoun, A., Saliba, N. A., Kushta, J., Christoudias, T., & Lakkis, I. (2022). Updated national emission inventory and comparison with the emissions database for global atmospheric research (EDGAR): Case of Lebanon. *Environmental Science and Pollution Research*, *29*(20), 30193–30205. https://doi.org/10.1007/s11356-021-17562-8

Silva, R. A., Adelman, Z., Fry, M. M., & West, J. J. (2016). The impact of individual anthropogenic emissions sectors on the global burden of human mortality due to ambient air pollution. *Environmental Health Perspectives*, *124*(11), 1776–1784. https://doi.org/10.1289/EHP177

Soulie, A., Granier, C., Darras, S., Zilbermann, N., Doumbia, T., Guevara, M., et al. (2024). Global anthropogenic emissions (CAMS-GLOB-ANT) for the copernicus atmosphere monitoring service simulations of air quality forecasts and reanalyses. *Earth System Science Data*, *16*(5), 2261–2279. https://doi.org/10.5194/essd-16-2261-2024

Soulie, A., Granier, C., Gon, H. D., Kuenen, J. J. P., Darras, S., Doumbia, T., et al. (2022). The copernicus atmosphere monitoring service (CAMS) emissions of greenhouse gases and air pollutants global anthropogenic emissions: Cams-glob-ant.

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Stanek, L. W., Brown, J. S., Stanek, J., Gift, J., & Costa, D. L. (2011). Air pollution toxicology—A brief review of the role of the science in shaping the current understanding of air pollution health risks. *Toxicological Sciences*, *120*(suppl_1), S8–S27. https://doi.org/10.1093/toxsci/kfq367

Stohl, A., Aamaas, B., Amann, M., Baker, L. H., Bellouin, N., Berntsen, T. K., et al. (2015). Evaluating the climate and air quality impacts of short-lived pollutants. *Atmospheric Chemistry and Physics*, *15*(18), 10529–10566. https://doi.org/10.5194/acp-15-10529-2015

Tang, W., Emmons, L. K., Buchholz, R. R., Wiedinmyer, C., Schwantes, R. H., He, C., et al. (2022). Effects of fire diurnal variation and plume rise on U.S. air quality during FIREX-AQ and WE-CAN based on the multi-scale infrastructure for chemistry and aerosols (MUSICAv0). *Journal of Geophysical Research: Atmospheres*, *127*(16), e2022JD036650. https://doi.org/10.1029/2022JD036650

Tang, W., Emmons, L. K., Worden, H. M., Kumar, R., He, C., Gaubert, B., et al. (2023). Application of the Multi-Scale Infrastructure for Chemistry and Aerosols version 0 (MUSICAv0) for air quality research in Africa. *Geoscientific Model Development*, *16*(20), 6001–6028. https://doi.org/10.5194/gmd-16-6001-2023

Thornhill, G. D., Collins, W. J., Kramer, R. J., Olivie, D., Skeie, R. B., O'Connor, F. M., et al. (2021). Effective radiative forcing from emissions of reactive gases and aerosols-A multi-model comparison. *Atmospheric Chemistry and Physics*, *21*(2), 853–874. https://doi.org/10.5194/acp-21-853-2021

Tilmes, S., Hodzic, A., Emmons, L. K., Mills, M. J., Gettelman, A., Kinnison, D. E., et al. (2019). Climate forcing and trends of organic aerosols in the community earth system model (CESM2). *Journal of Advances in Modeling Earth Systems*, *11*(12), 4323–4351. https://doi.org/10.1029/2019MS001827

Tilmes, S., Lamarque, J. F., Emmons, L. K., Kinnison, D. E., Ma, P. L., Liu, X., et al. (2015). Description and evaluation of tropospheric chemistry and aerosols in the community earth system model (CESM1.2). *Geoscientific Model Development*, *8*(5), 1395–1426. https://doi.org/10.5194/gmd-8-1395-2015

Tsigaridis, K., Daskalakis, N., Kanakidou, M., Adams, P. J., Artaxo, P., Bahadur, R., et al. (2014). The AeroCom evaluation and intercomparison of organic aerosol in global models. *Atmospheric Chemistry and Physics*, *14*(19), 10845–10895. https://doi.org/10.5194/acp-14-10845-2014

Turner, M. C., Jerrett, M., Pope, C. A., Krewski, D., Gapstur, S. M., Diver, W. R., et al. (2016). Long-term ozone exposure and mortality in a large prospective study. *American Journal of Respiratory and Critical Care Medicine*, *193*(10), 1134–1142. https://doi.org/10.1164/rccm.201508-1633OC

Upadhyay, A., Dey, S., & Goyal, P. (2020). A comparative assessment of regional representativeness of EDGAR and ECLIPSE emission inventories for air quality studies in India. *Atmospheric Environment*, *223*, 117182. https://doi.org/10.1016/j.atmosenv.2019.117182

Van Donkelaar, A., Hammer, M. S., Bindle, L., Brauer, M., Brook, J. R., Garay, M. J., et al. (2021). Monthly global estimates of fine particulate matter and their uncertainty. *Environmental Science and Technology*, *55*(22), 15287–15300. https://doi.org/10.1021/acs.est.1c05309

Van Donkelaar, A., Martin, R. V., Brauer, M., Hsu, N. C., Kahn, R. A., Levy, R. C., et al. (2016). Global estimates of fine particulate matter using a combined geophysical-statistical method with information from satellites, models, and monitors. *Environmental Science and Technology*, *50*(7), 3762–3772. https://doi.org/10.1021/acs.est.5b05833

Van Marle, M. J. E., Kloster, S., Magi, B. I., Marlon, J. R., Daniau, A. L., Field, R. D., et al. (2017). Historic global biomass burning emissions for CMIP6 (BB4CMIP) based on merging satellite observations with proxies and fire models (1750-2015). *Geoscientific Model Development*, *10*(9), 3329–3357. https://doi.org/10.5194/gmd-10-3329-2017

Wang, Q., Wang, J., He, M. Z., Kinney, P. L., & Li, T. (2018). A county-level estimate of PM2.5 related chronic mortality risk in China based on multi-model exposure data. *Environment International*, *110*(February), 105–112. https://doi.org/10.1016/j.envint.2017.10.015

World Health Organization. (2024). Air pollution.

Wu, M., Liu, X., Yang, K., Luo, T., Wang, Z., Wu, C., et al. (2019). Modeling dust in east Asia by CESM and sources of biases. *Journal of Geophysical Research: Atmospheres*, *124*(14), 8043–8064. https://doi.org/10.1029/2019JD030799

Xiong, Y., Du, K., & Huang, Y. (2024). One-third of global population at cancer risk due to elevated volatile organic compounds levels. *Npj Climate and Atmospheric Science*, *7*(1), 54. https://doi.org/10.1038/s41612-024-00598-1

Xiong, Y., Huang, Y., & Du, K. (2022). Health risk-oriented source apportionment of hazardous volatile organic compounds in eight Canadian cities and implications for prioritizing mitigation strategies. *Environmental Science and Technology*, *56*(17), 12077–12085. https://doi.org/10.1021/acs.est.2c02558

Xiong, Y., Partha, D., Prime, N., Smith, S. J., Mariscal, N., Salah, H., & Huang, Y. (2022). Long-term trends of impacts of global gasoline and diesel emissions on ambient PM2.5 and O3 pollution and the related health burden for 2000-2015. *Environmental Research Letters*, *17*(10), 104042. https://doi.org/10.1088/1748-9326/ac9422

Young, P. J., Archibald, A. T., Bowman, K. W., Lamarque, J.-F., Naik, V., Stevenson, D. S., et al. (2013). Pre-industrial to end 21st century projections of tropospheric ozone from the atmospheric chemistry and climate model intercomparison project (ACCMIP). *Atmospheric Chemistry and Physics*, *13*(4), 2063–2090. https://doi.org/10.5194/acp-13-2063-2013

Young, P. J., Naik, V., Fiore, A. M., Gaudel, A., Guo, J., Lin, M. Y., et al. (2018). *Tropospheric ozone assessment report: Assessment of global-scale model performance for global and regional ozone distributions, variability, and trends*. Elementa. University of California Press. https://doi.org/10.1525/elementa.265

Zender, C. S., Bian, H., & Newman, D. (2003). Mineral dust entrainment and deposition (DEAD) model: Description and 1990s dust climatology. *Journal of Geophysical Research*, *108*(14). https://doi.org/10.1029/2002jd002775

Zhai, S., Jacob, D. J., Wang, X., Shen, L., Li, K., Zhang, Y., et al. (2019). Fine particulate matter (PM2.5) trends in China, 2013-2018. Separating contributions from anthropogenic emissions and meteorology. *Atmospheric Chemistry and Physics*, *19*(16), 11031–11041. https://doi.org/10.5194/acp-19-11031-2019

Zhang, X., Cheng, C., & Zhao, H. (2022). A health impact and economic loss assessment of O3 and PM2.5 exposure in China from 2015 to 2020. *GeoHealth*, *6*(3), 1–15. https://doi.org/10.1029/2021GH000531

Zhao, H., Li, X., Zhang, Q., Jiang, X., Lin, J., Peters, G. P., et al. (2017). Effects of atmospheric transport and trade on air pollution mortality in China. *Atmospheric Chemistry and Physics*, *17*(16), 10367–10381. https://doi.org/10.5194/acp-17-10367-2017

References From the Supporting Information

Flato, G. M. (2011). Earth system models: An overview. *Wiley Interdisciplinary Reviews: Climate Change*, *2*(6), 783–800. https://doi.org/10.1002/wcc.148

Lamarque, J. F., Bond, T. C., Eyring, V., Granier, C., Heil, A., Klimont, Z., et al. (2010). Historical (1850-2000) gridded anthropogenic and biomass burning emissions of reactive gases and aerosols: Methodology and application. *Atmospheric Chemistry and Physics*, *10*(15), 7017–7039. https://doi.org/10.5194/acp-10-7017-2010

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